

Related U.S. Application Data			
division of application No. 16/928,983, filed on Jul. 14, 2020, now Pat. No. 11,594,001.			
(60)	Provisional application No. 62/963,494, filed on Jan. 20, 2020.	8,559,592 B2	10/2013 Betcke
(51)	Int. Cl. <i>G01N 23/083</i> (2018.01) <i>G01N 23/10</i> (2018.01) <i>G06F 3/16</i> (2006.01) <i>G06T 7/00</i> (2017.01) <i>G06T 11/00</i> (2006.01) <i>G06T 13/20</i> (2011.01)	8,625,735 B2	1/2014 Morton
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(52)	U.S. Cl. CPC <i>G06F 3/165</i> (2013.01); <i>G06T 7/0002</i> (2013.01); <i>G06T 11/008</i> (2013.01); <i>G06T 13/20</i> (2013.01); <i>G01N 2223/3307</i> (2013.01); <i>G01N 2223/401</i> (2013.01); <i>G01N 2223/421</i> (2013.01); <i>G01N 2223/639</i> (2013.01); <i>G01N 2223/643</i> (2013.01); <i>G06T 2200/24</i> (2013.01); <i>G06T 2219/2012</i> (2013.01)	9,046,465 B2	6/2015 Thompson
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(58)	Field of Classification Search CPC ... <i>G06T 2219/2012</i> ; <i>G06T 2207/10081</i> ; <i>G06T 2207/30112</i> ; <i>G06T 2219/004</i> ; <i>G06T 7/0004</i> ; <i>G01N 23/04</i> ; <i>G01N 23/083</i> ; <i>G01N 23/10</i> ; <i>G01N 2223/3307</i> ; <i>G01N 2223/401</i> ; <i>G01N 2223/421</i> ; <i>G01N 2223/639</i> ; <i>G01N 2223/643</i> ; <i>G01N 2223/501</i> ; <i>G01N 23/046</i> ; <i>G06F 3/165</i> See application file for complete search history.	9,420,677 B2	8/2016 Morton
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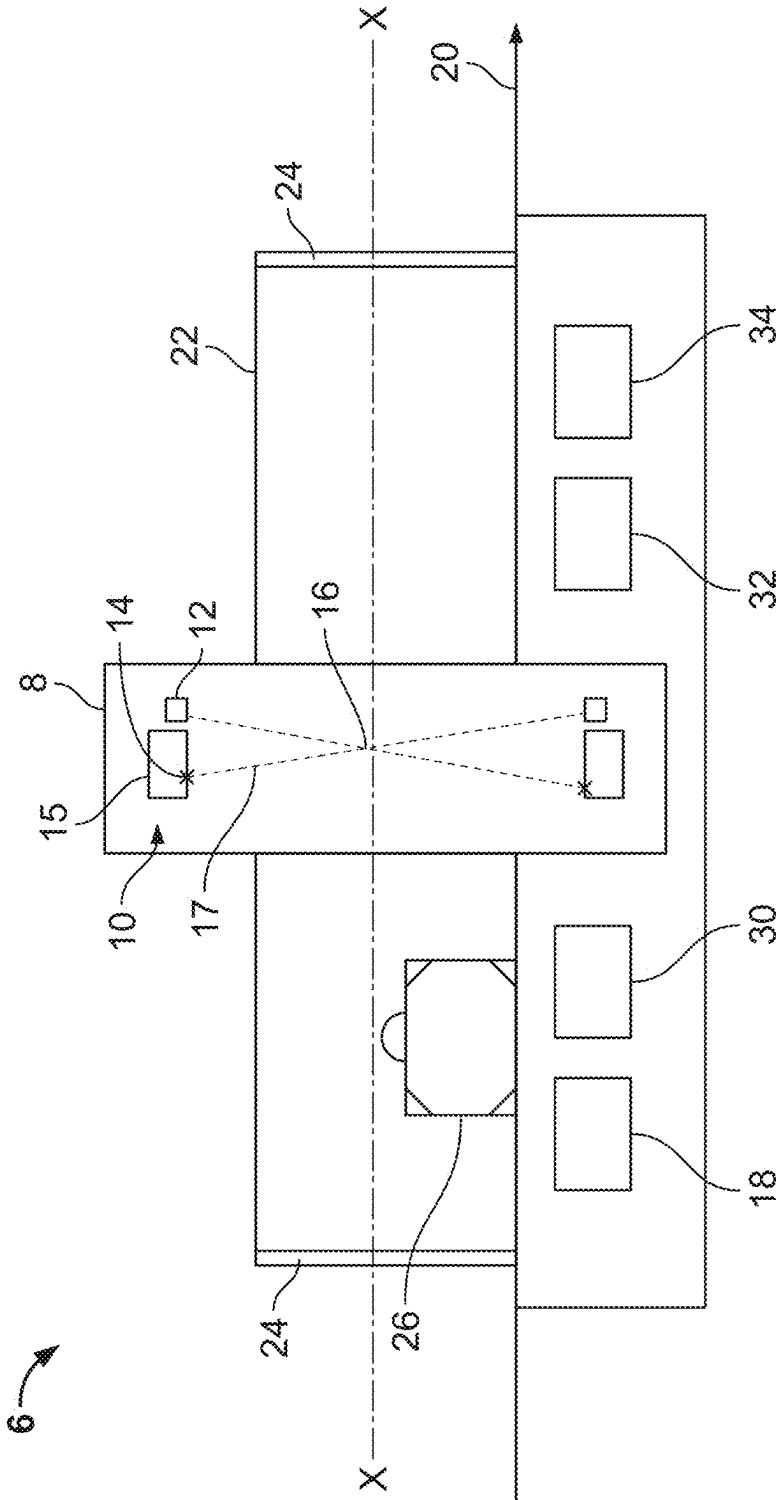


FIG. 1

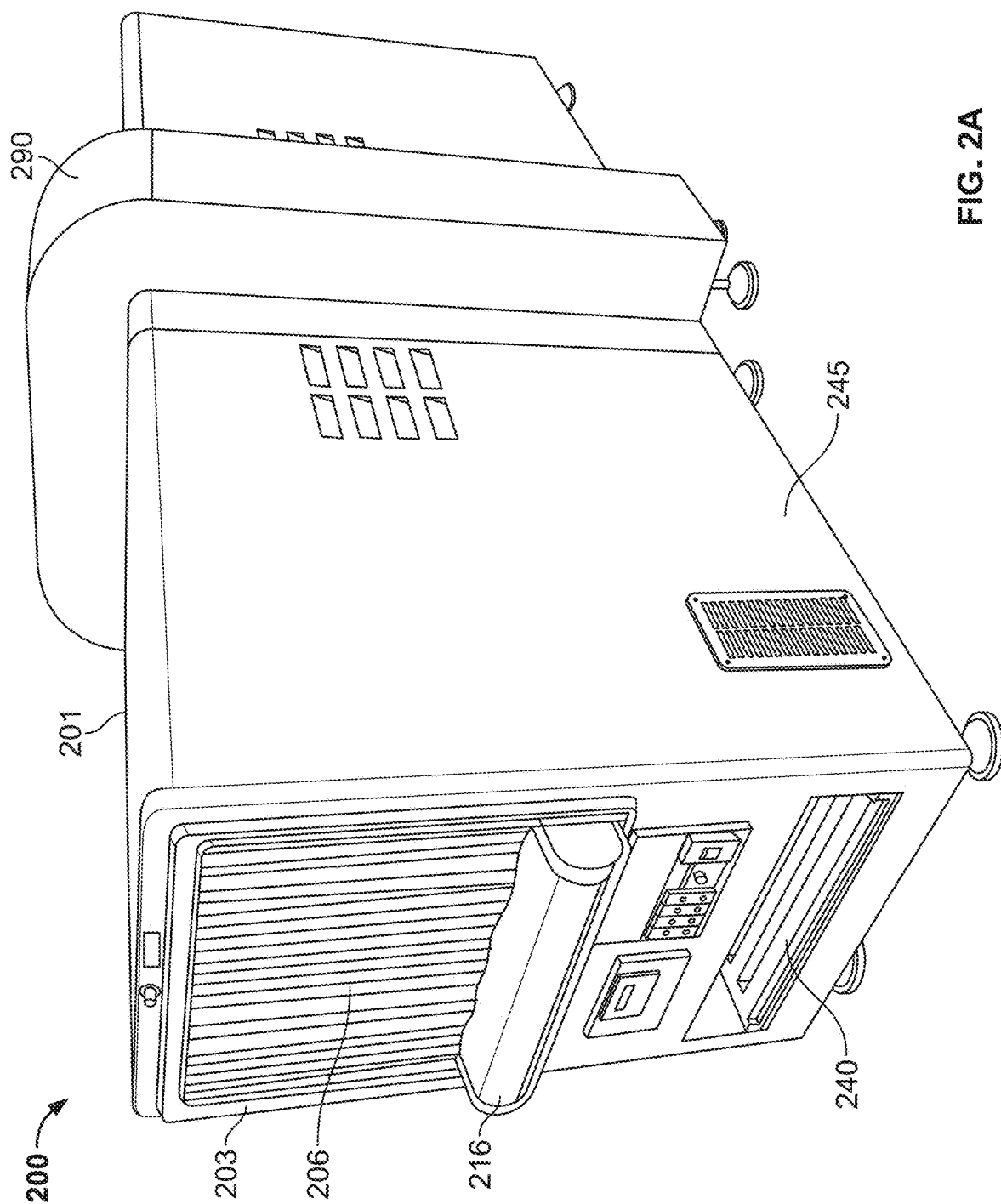
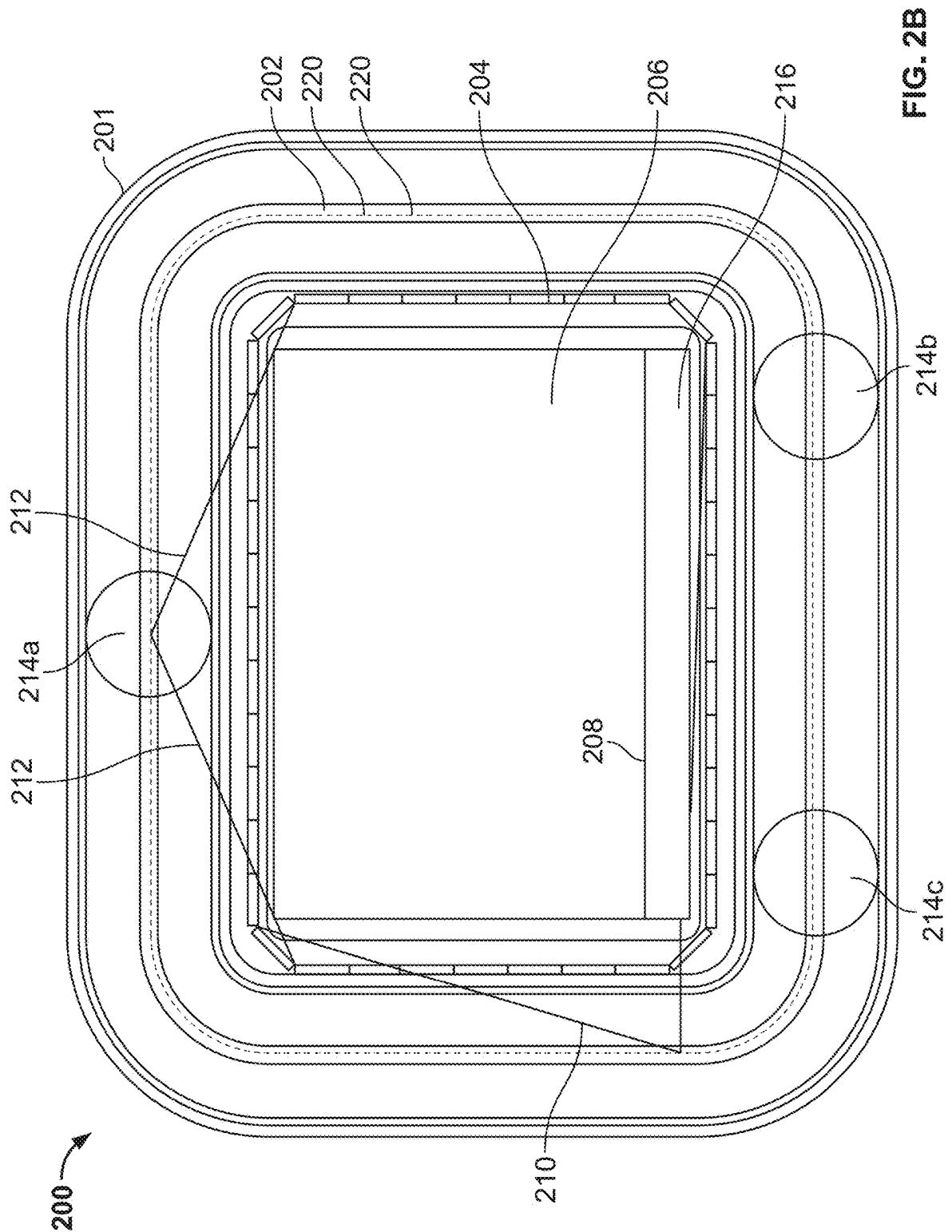


FIG. 2A



300

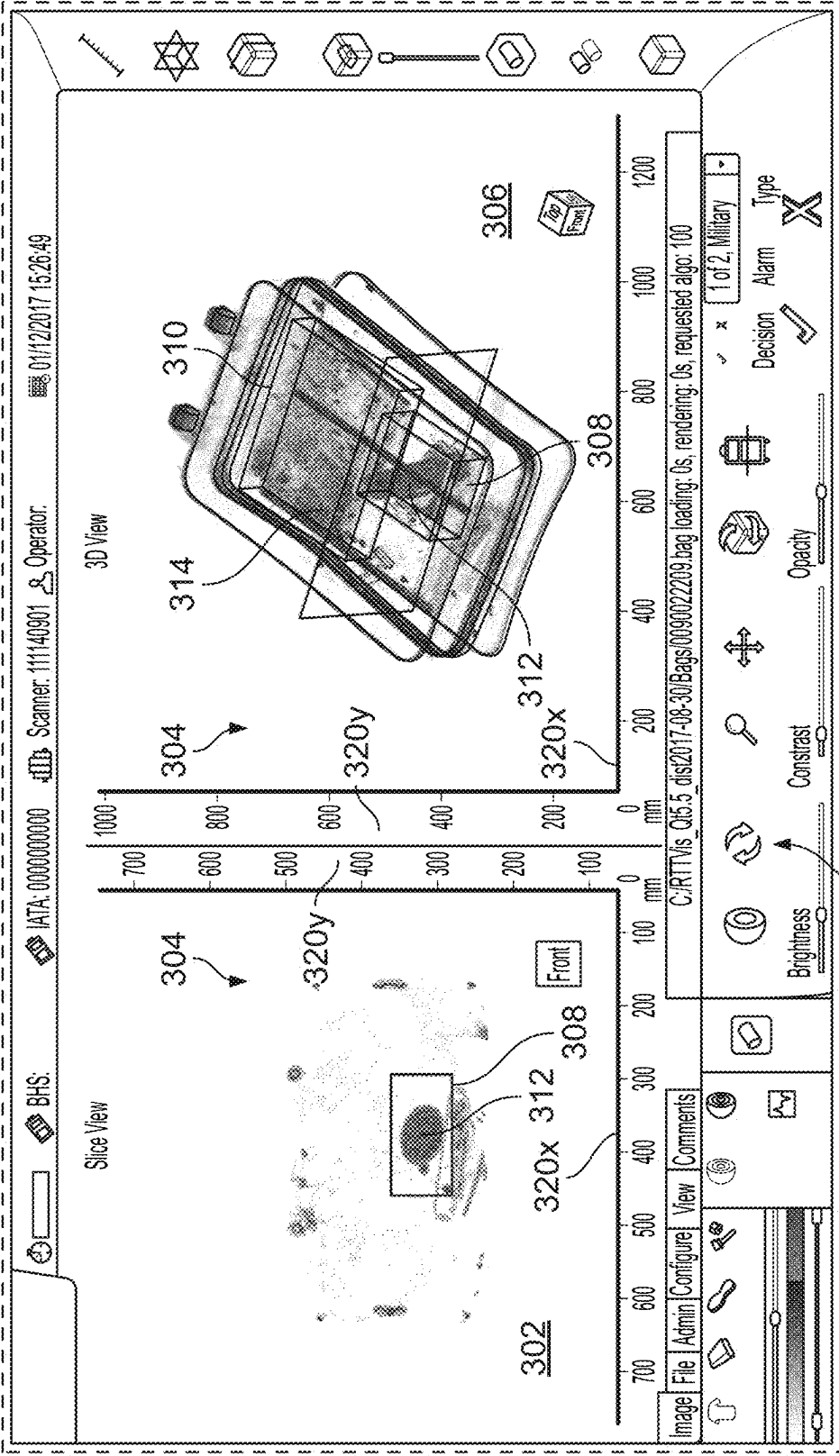


FIG. 3

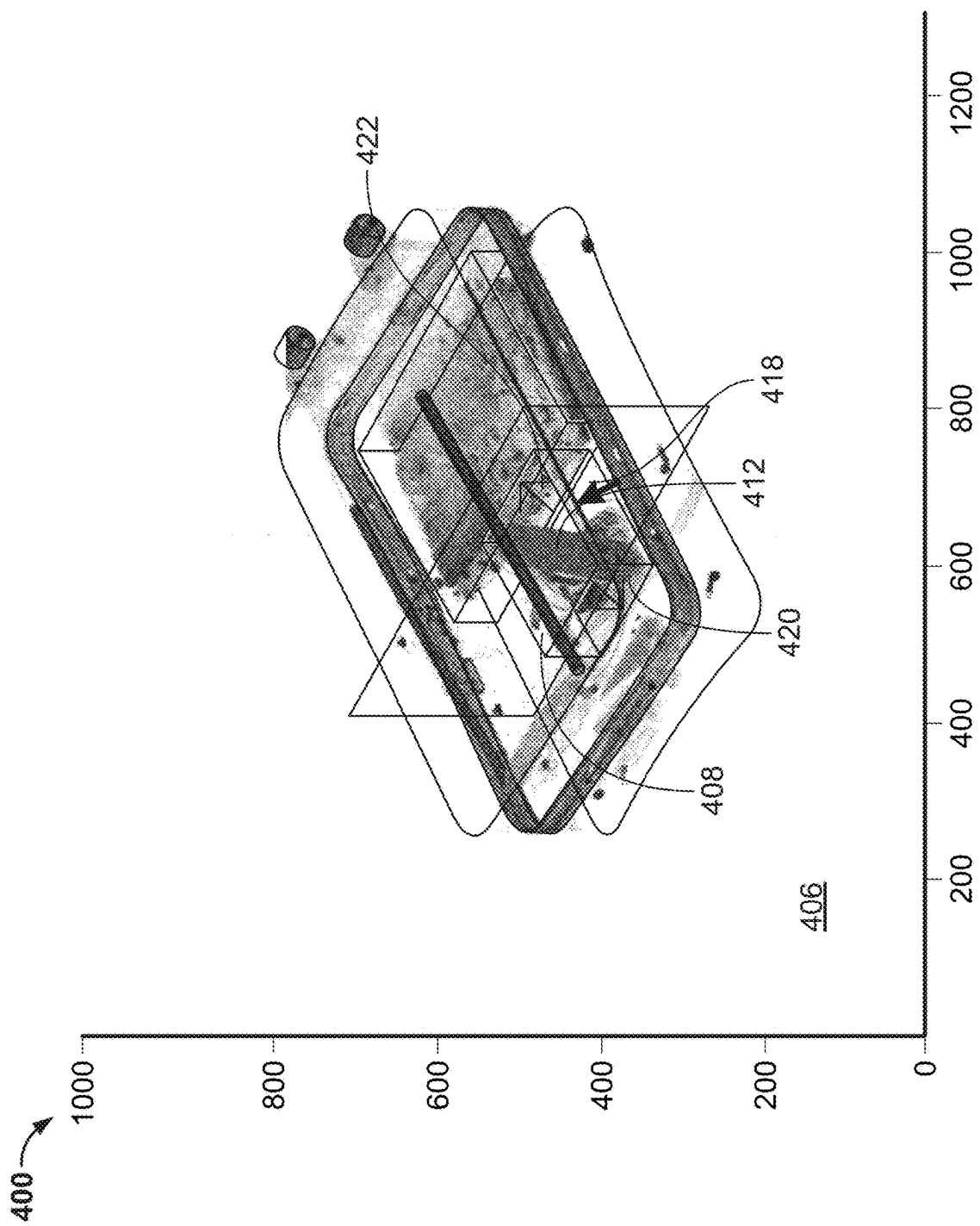


FIG. 4A

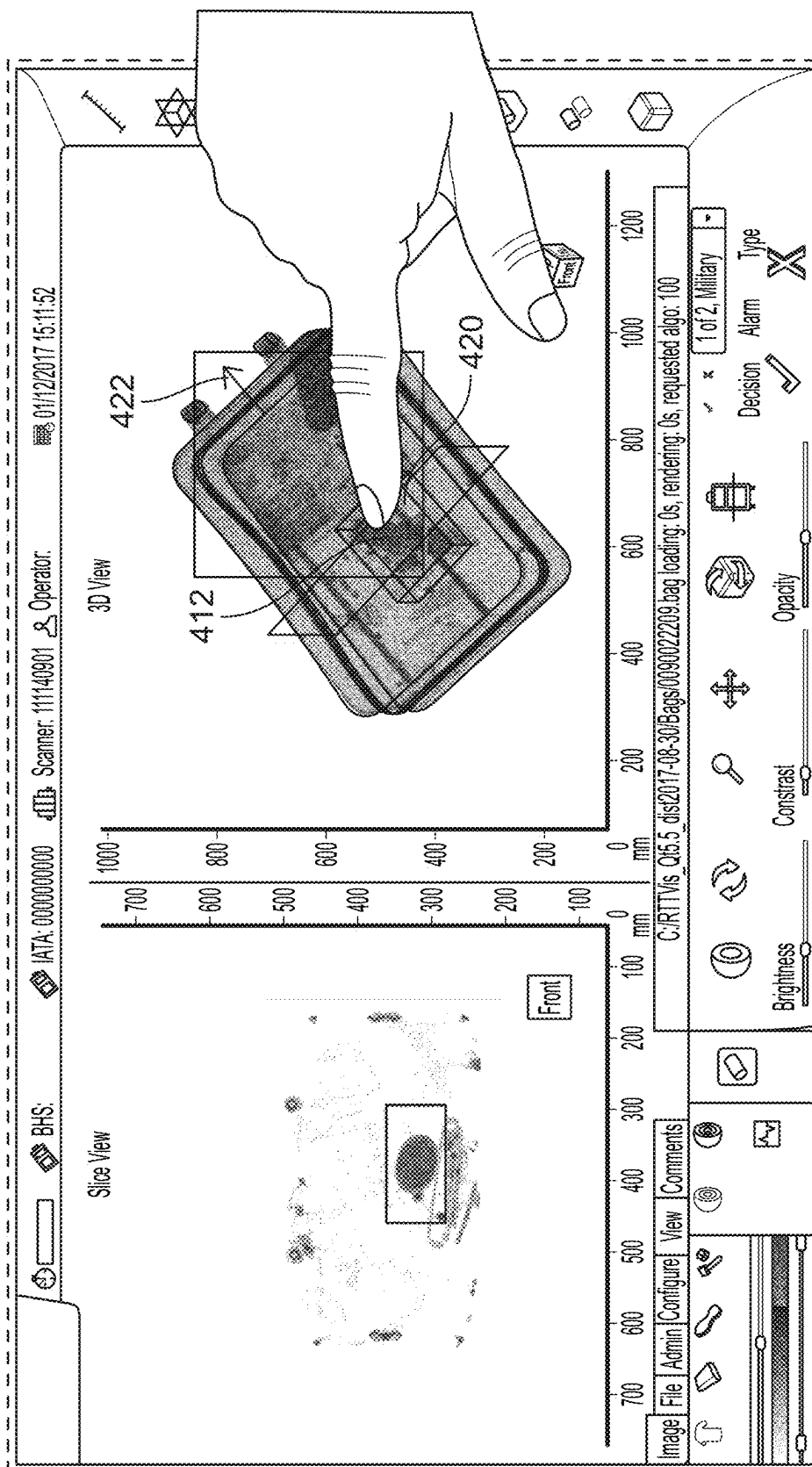


FIG. 4B

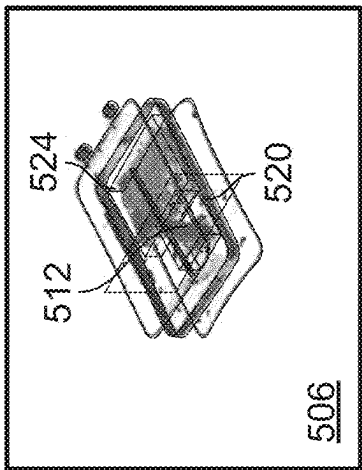


FIG. 5A

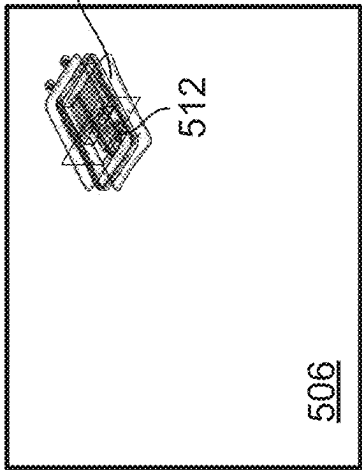


FIG. 5B

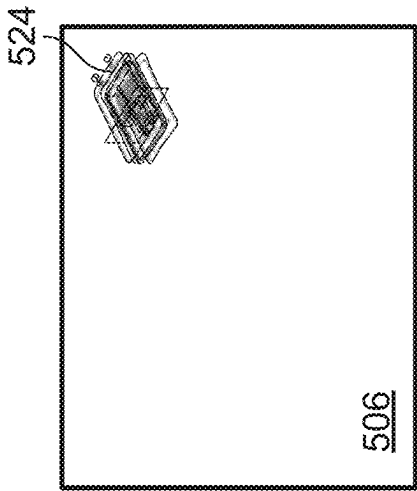


FIG. 5C

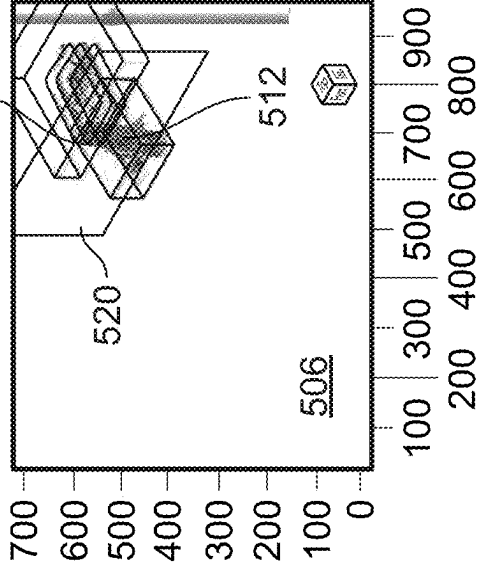


FIG. 5D

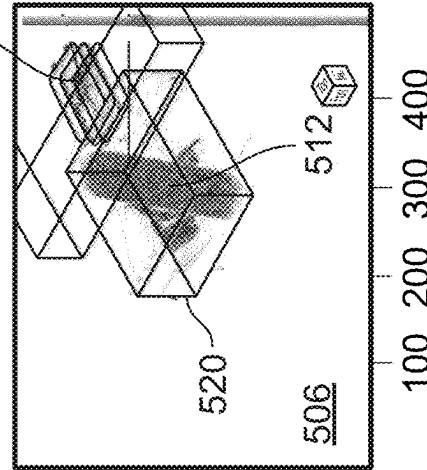


FIG. 5E

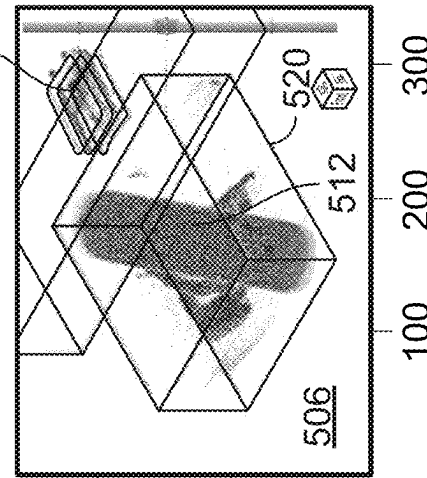


FIG. 5F

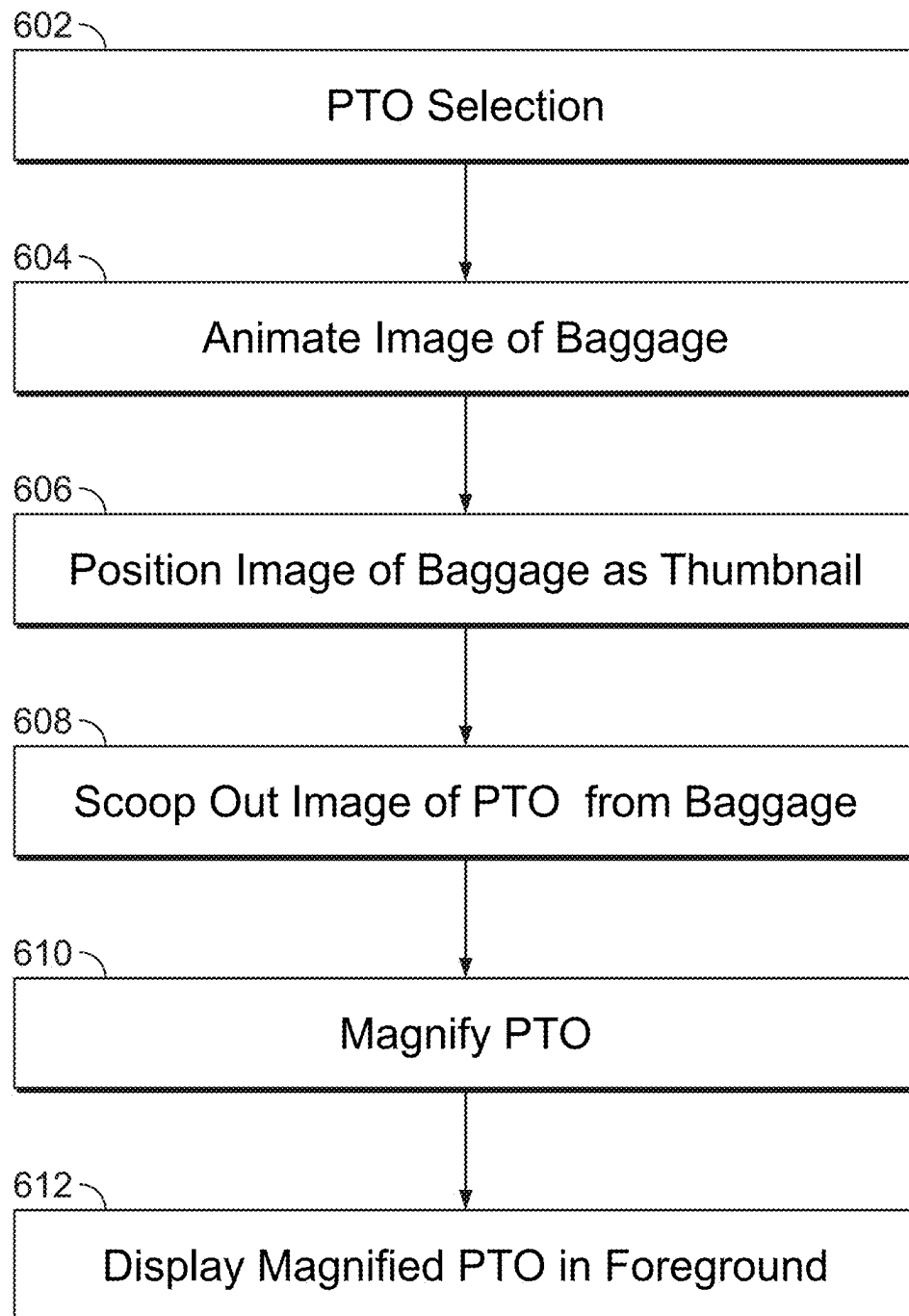
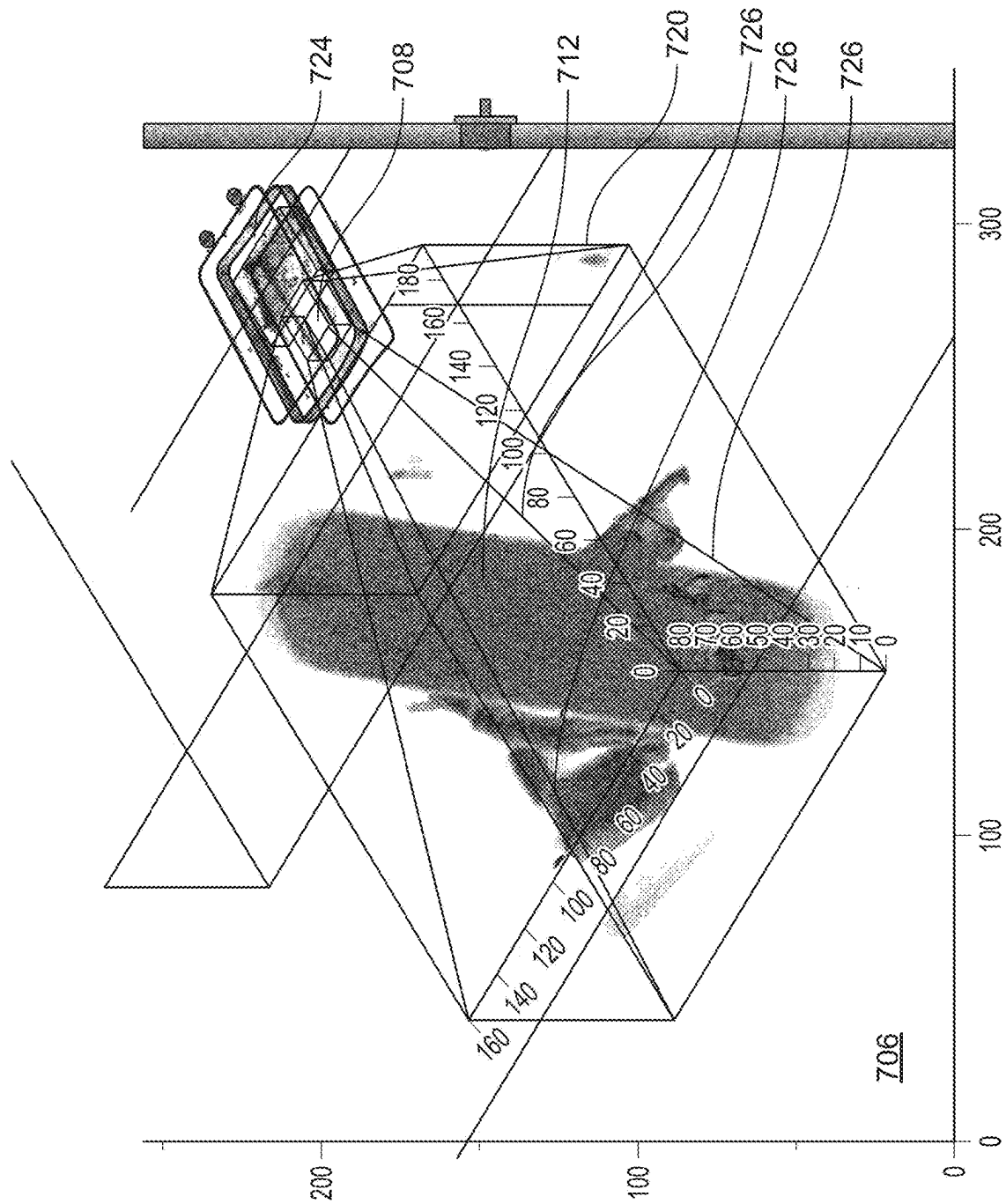
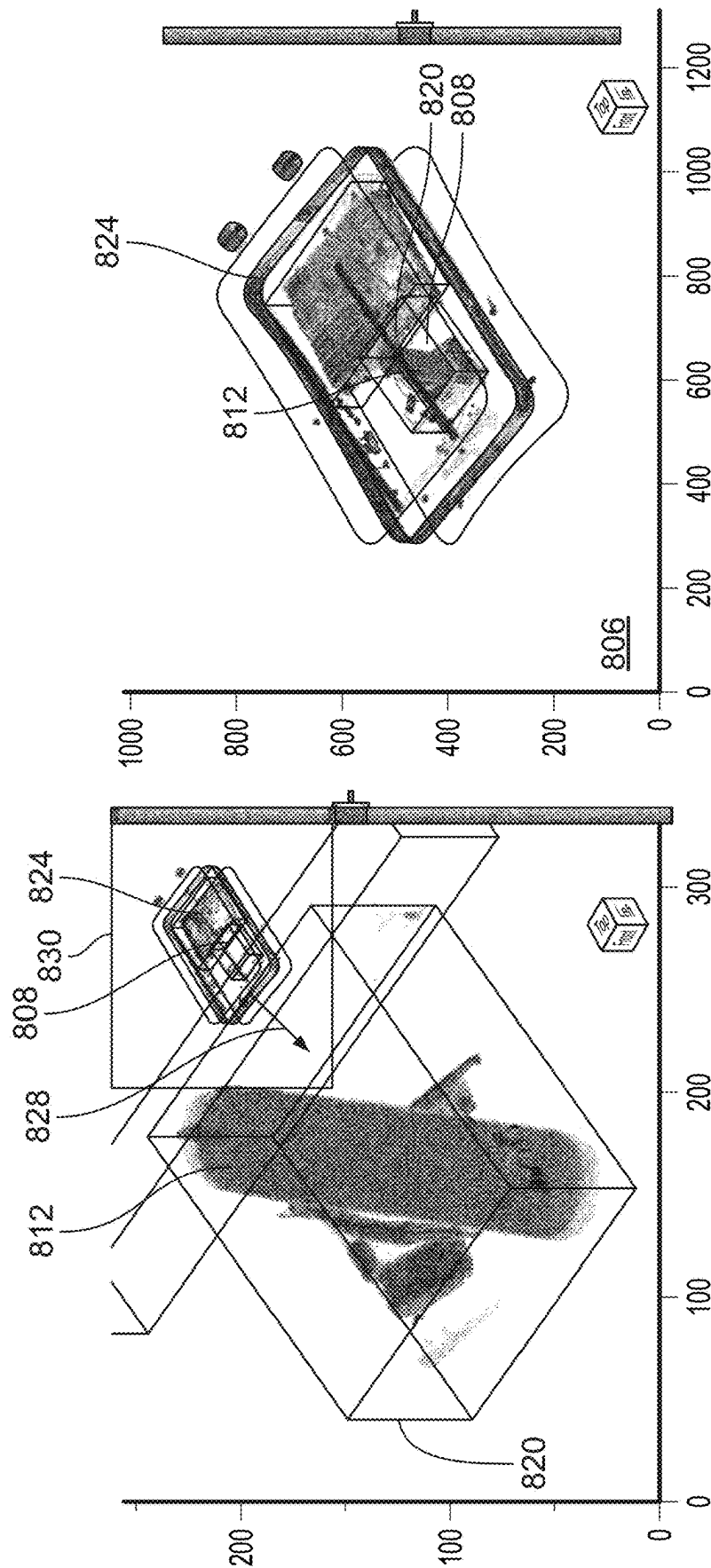
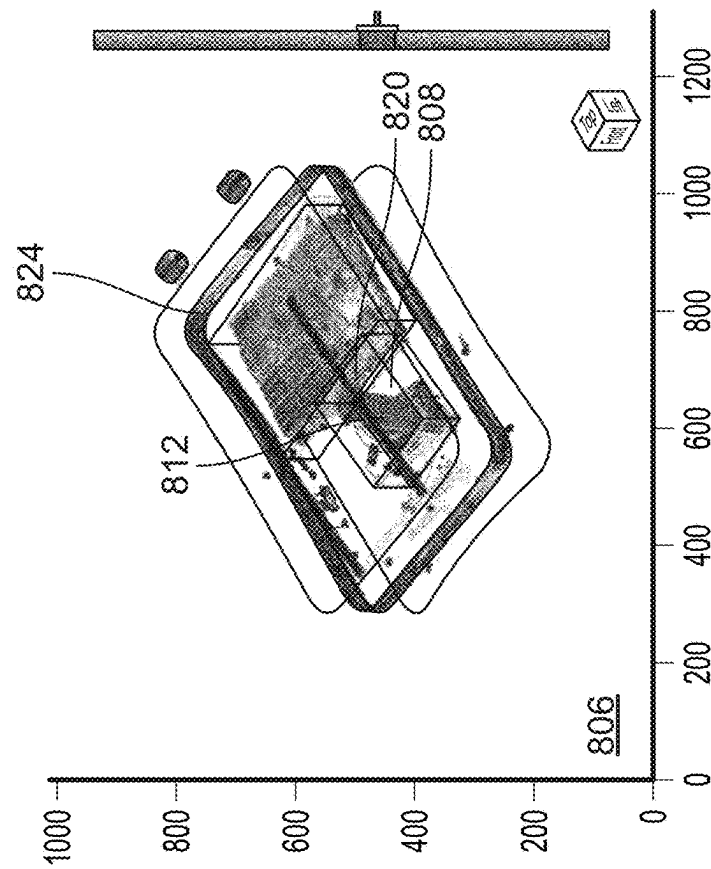


FIG. 6





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 DEPARTMENT OF HEALTH AND HUMAN SERVICES

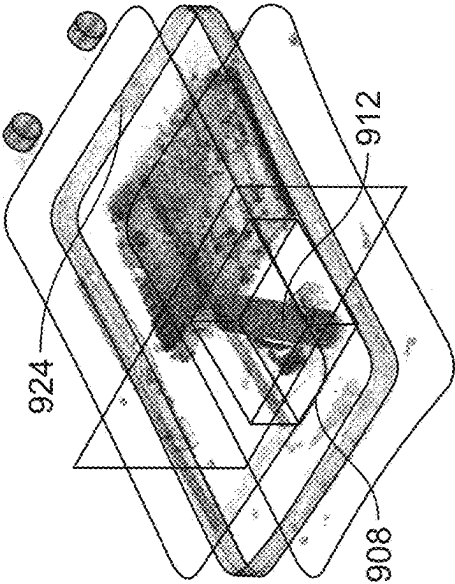


FIG. 9A

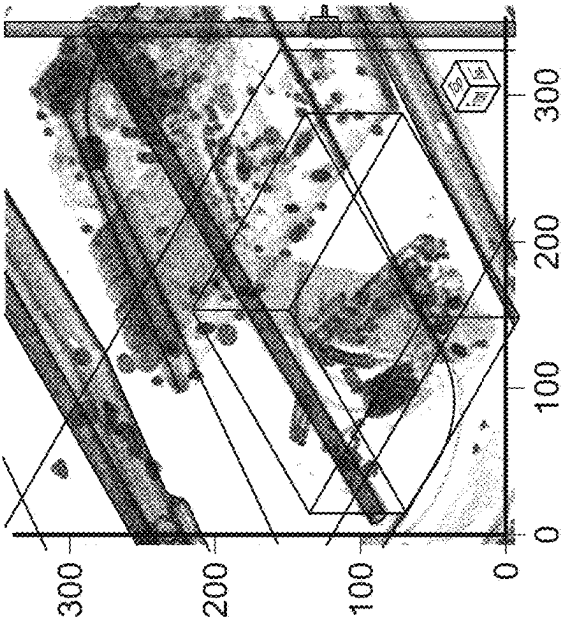


FIG. 9B

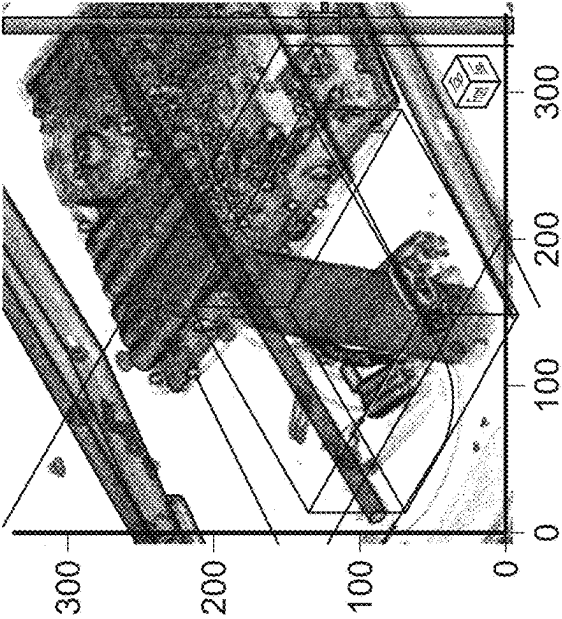


FIG. 9C

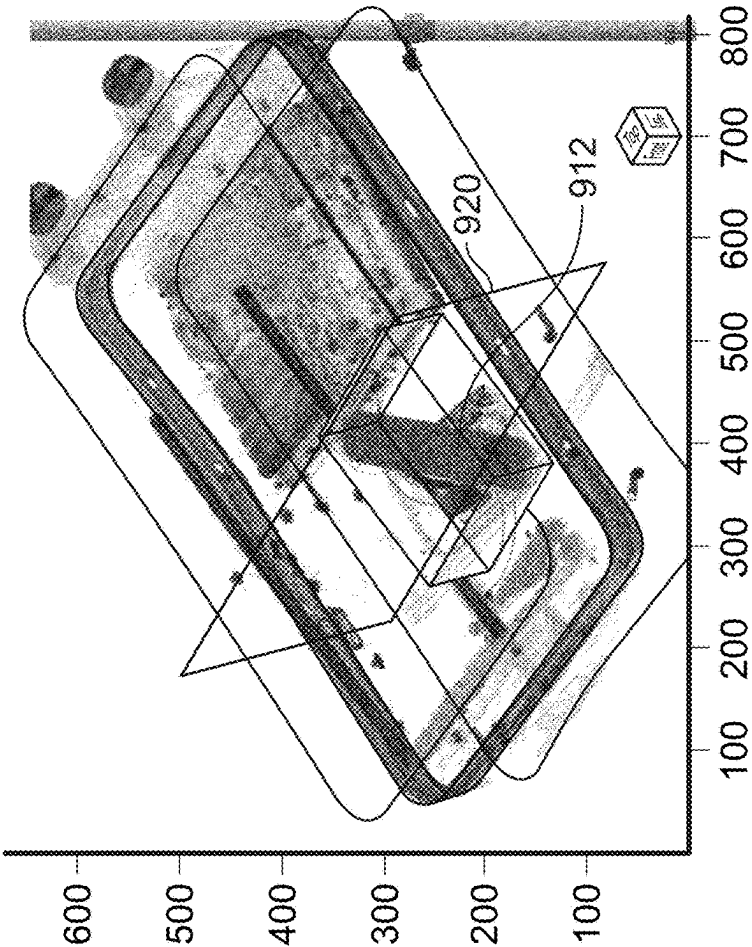


FIG. 9D

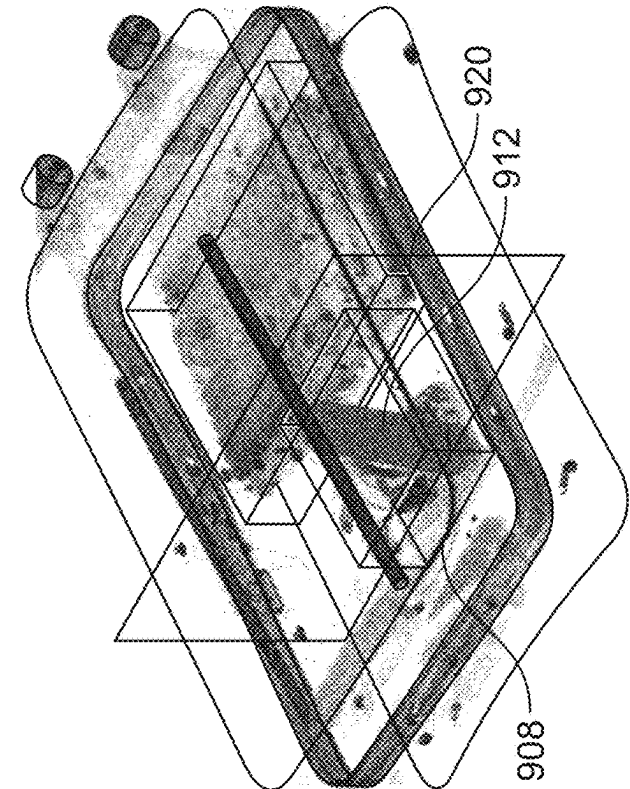


FIG. 9E

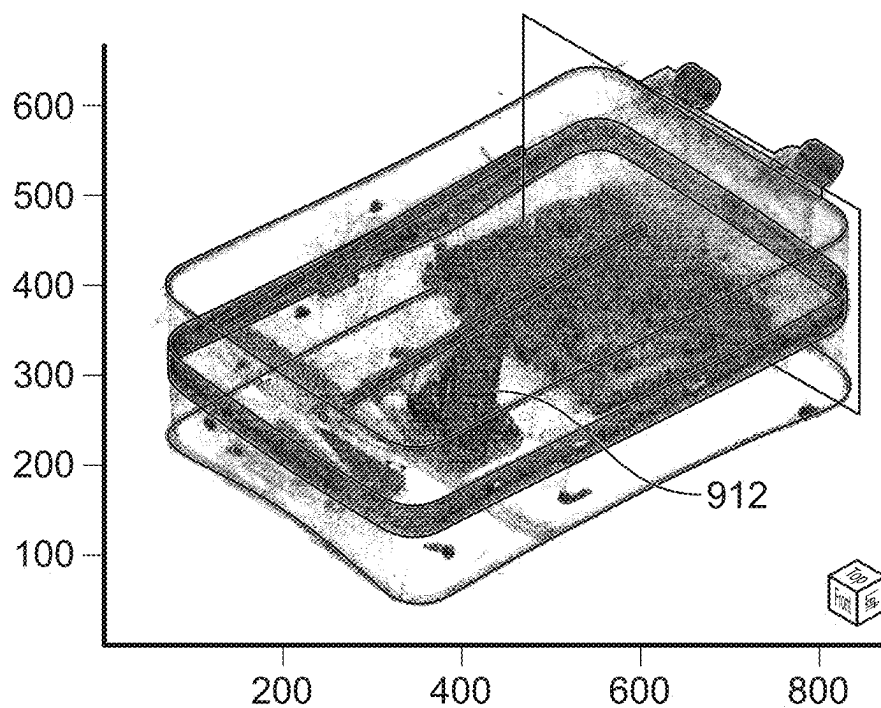


FIG. 9F

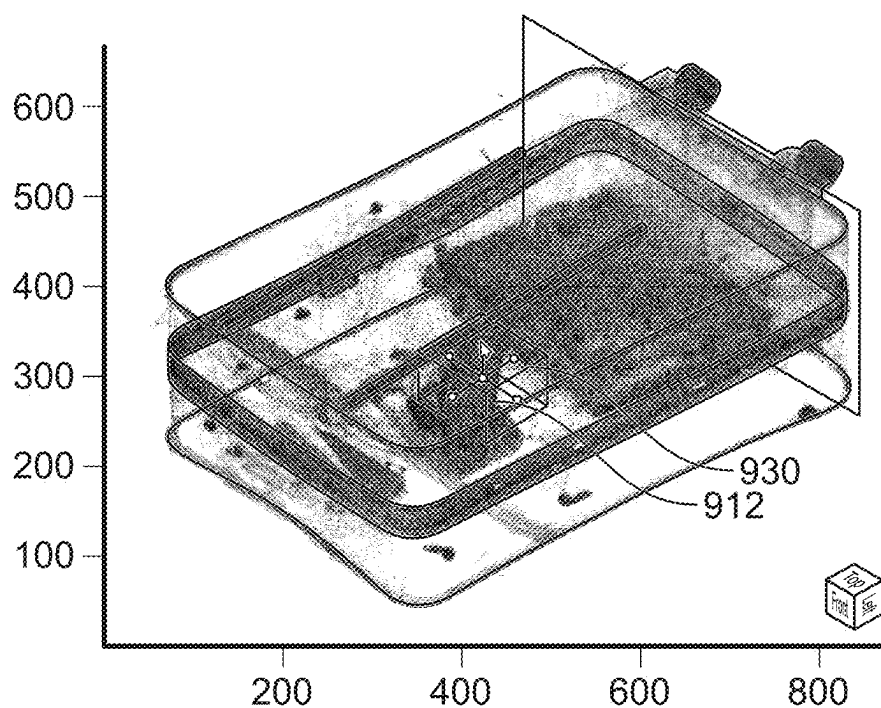


FIG. 9G

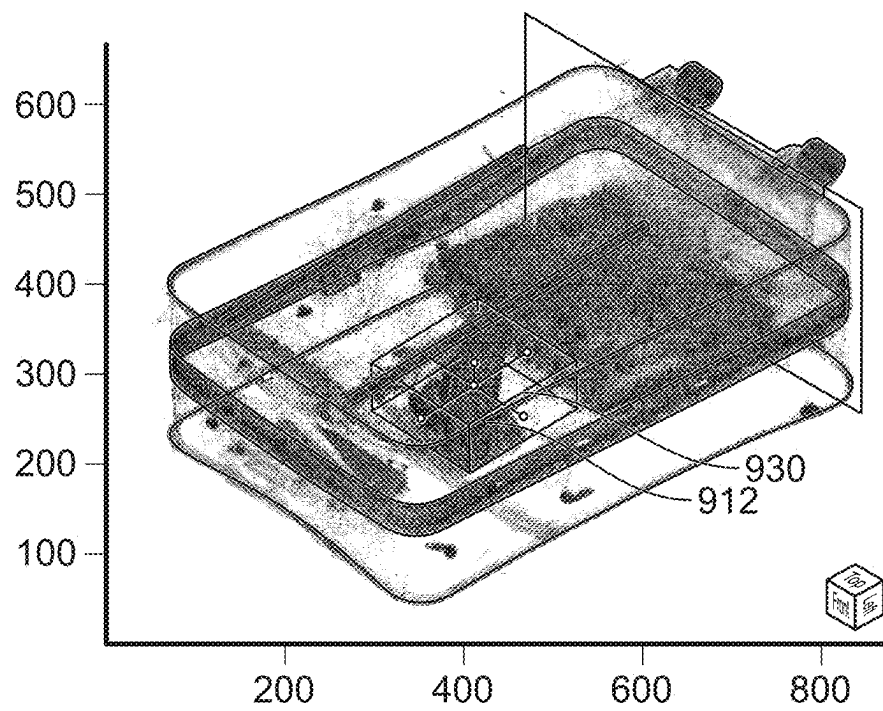


FIG. 9H

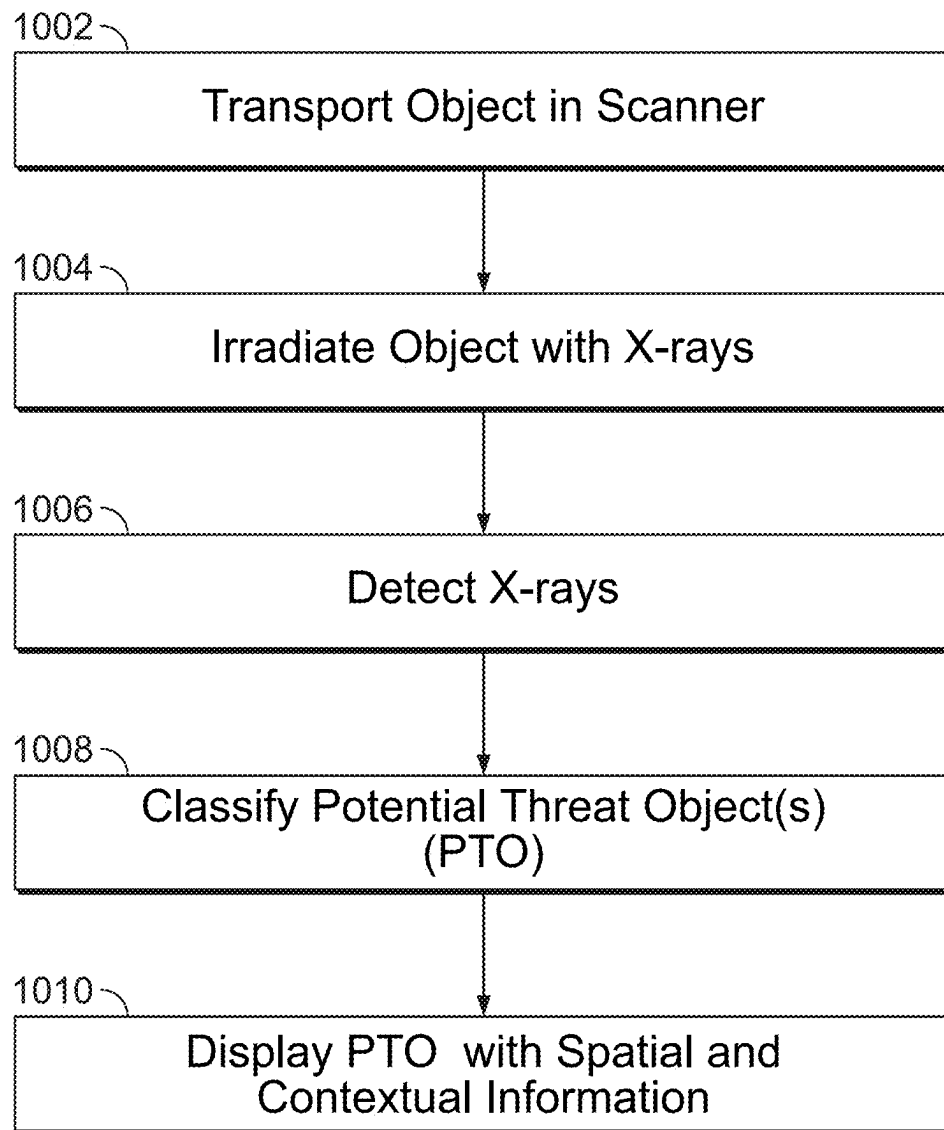


FIG. 10

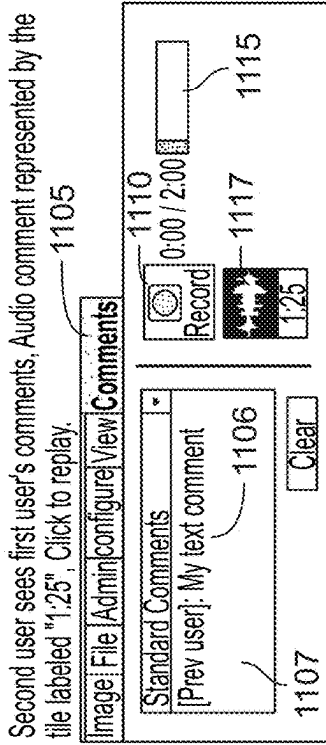


FIG. 11D

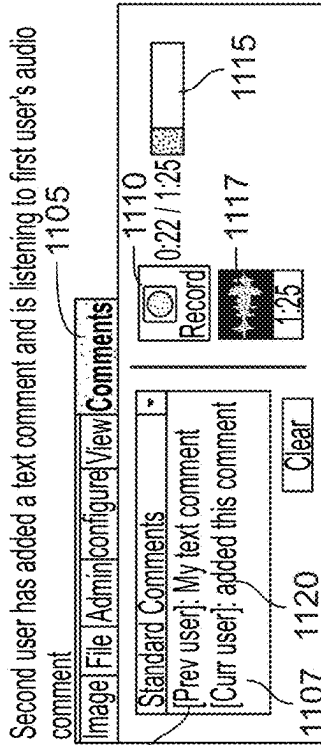


FIG. 11E

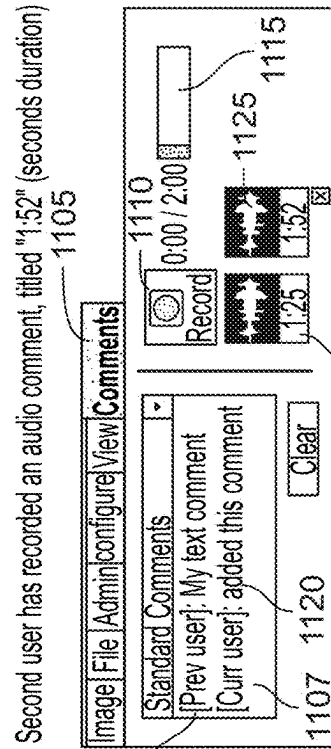


FIG. 11F

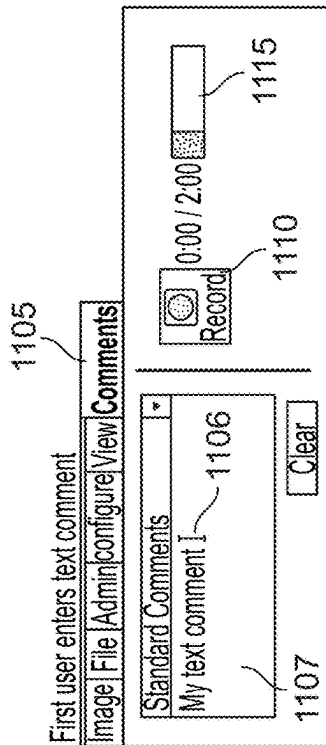


FIG. 11A

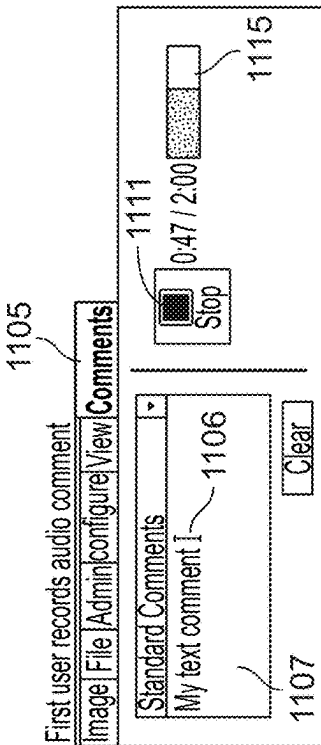


FIG. 11B

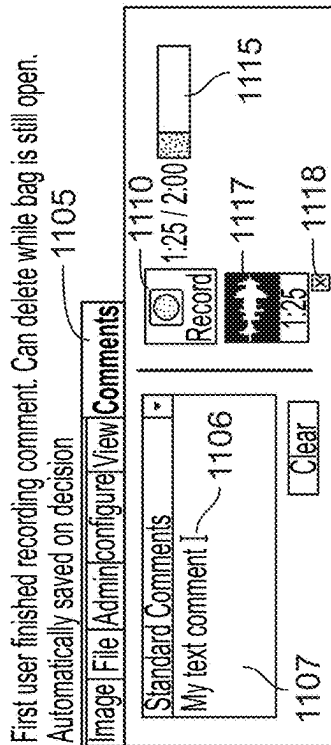


FIG. 11C

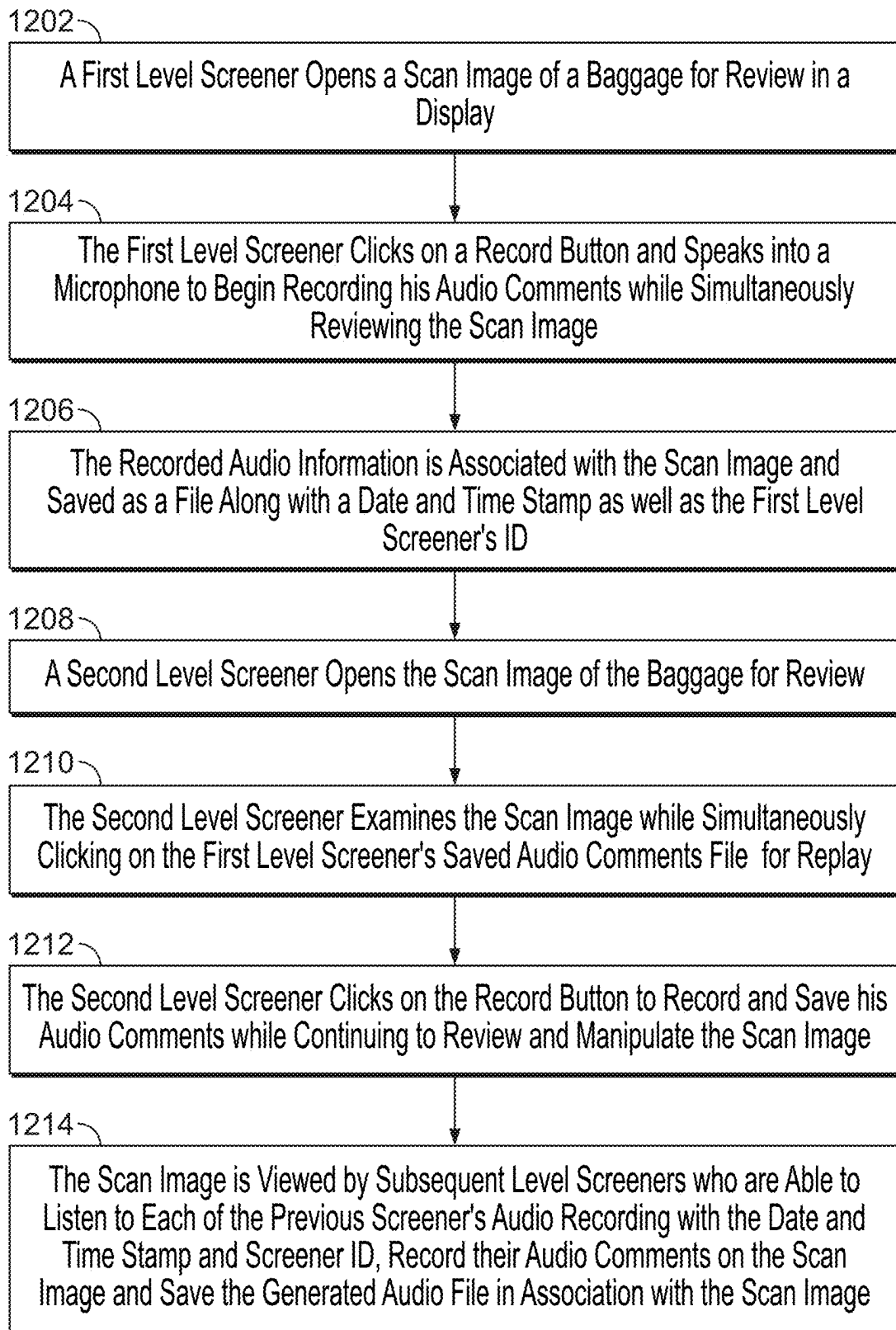
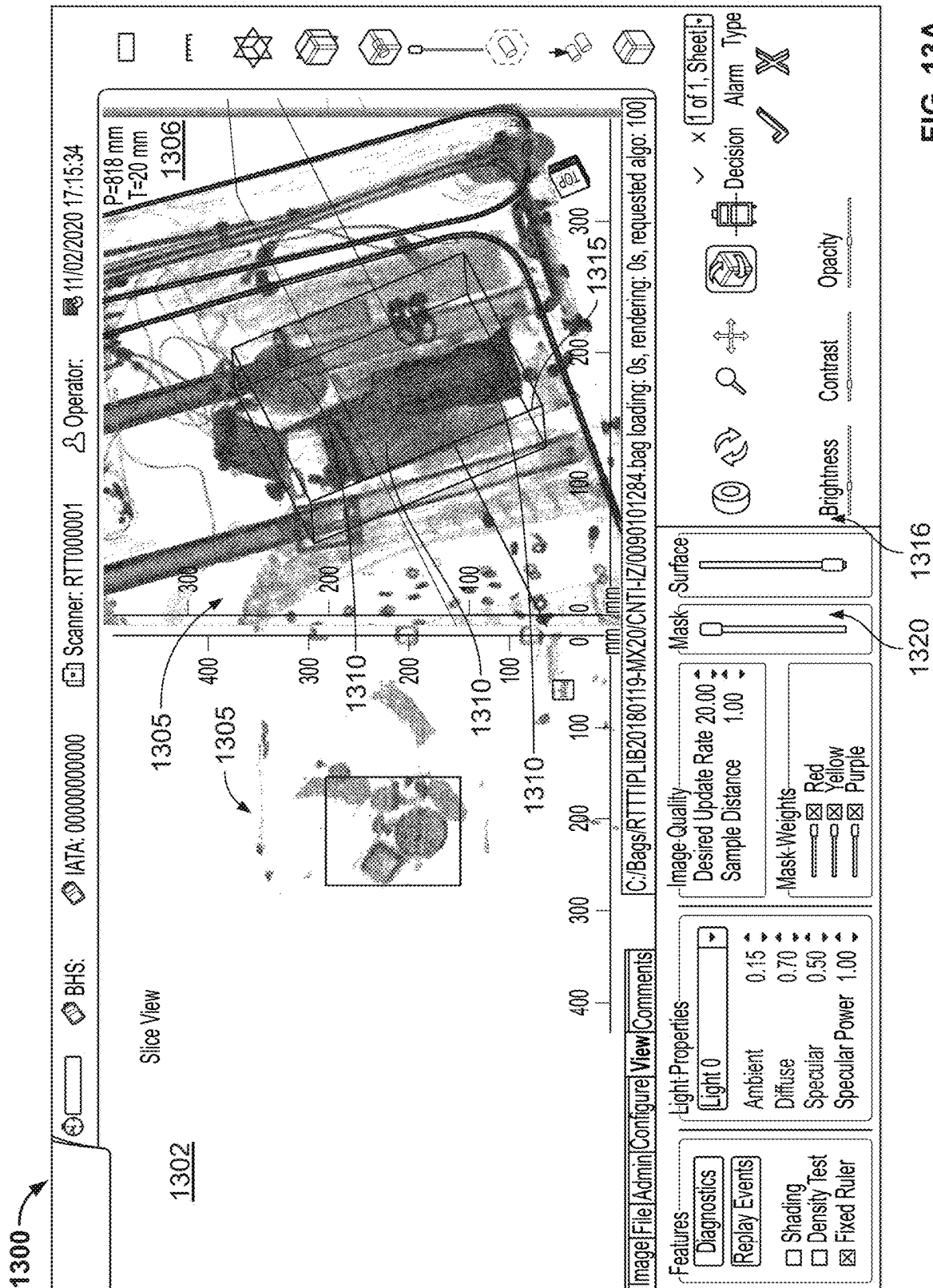
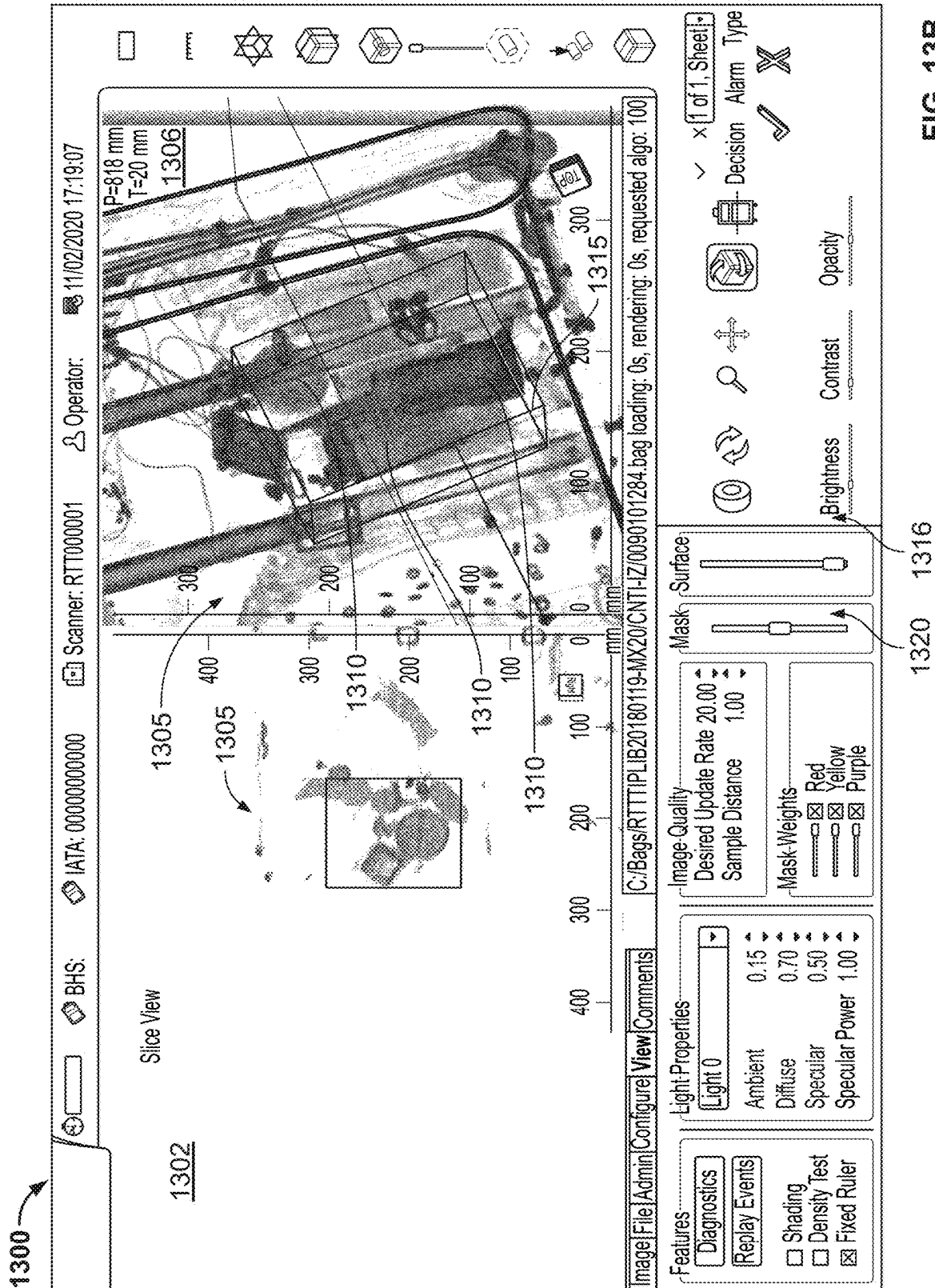
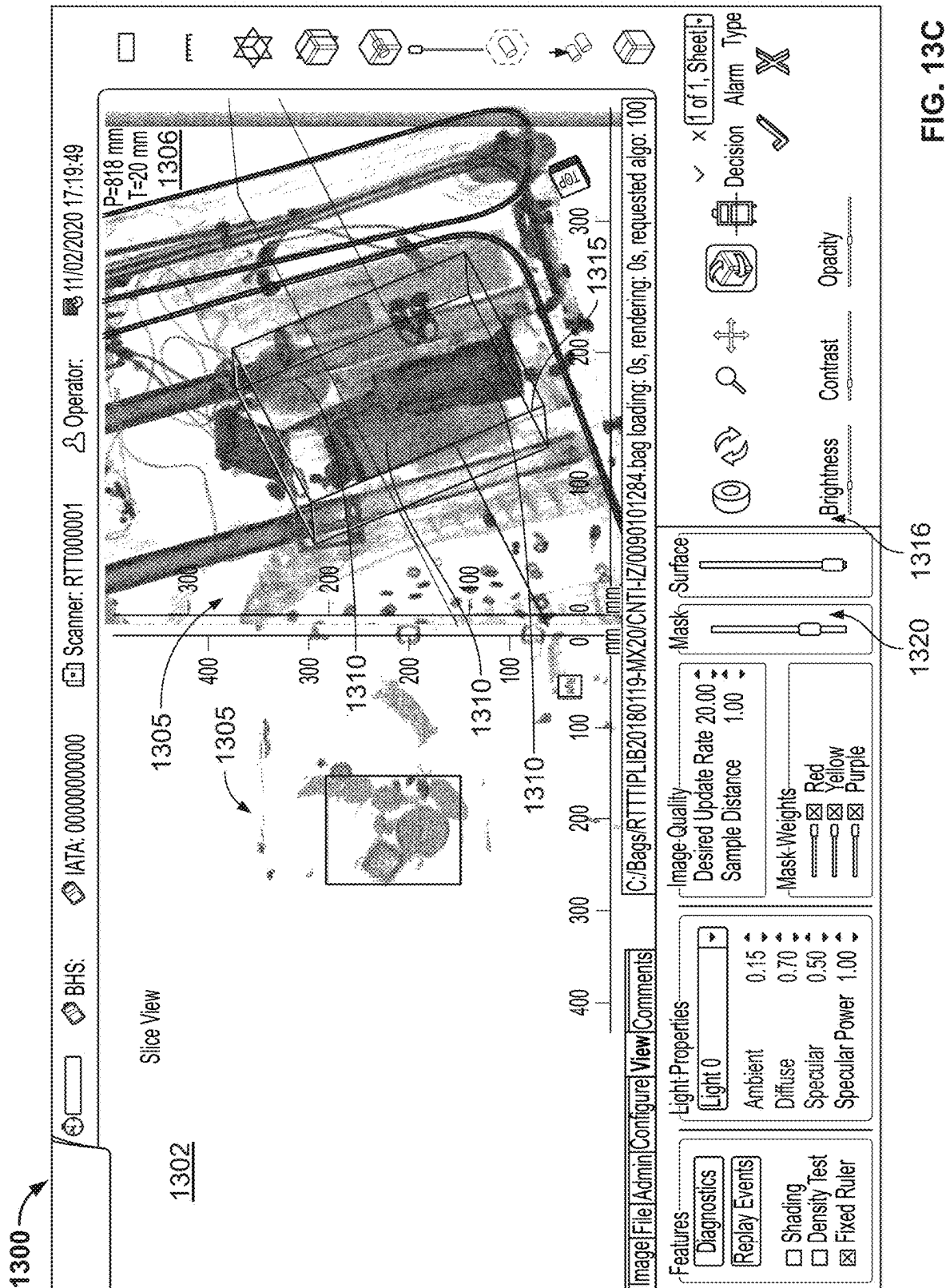
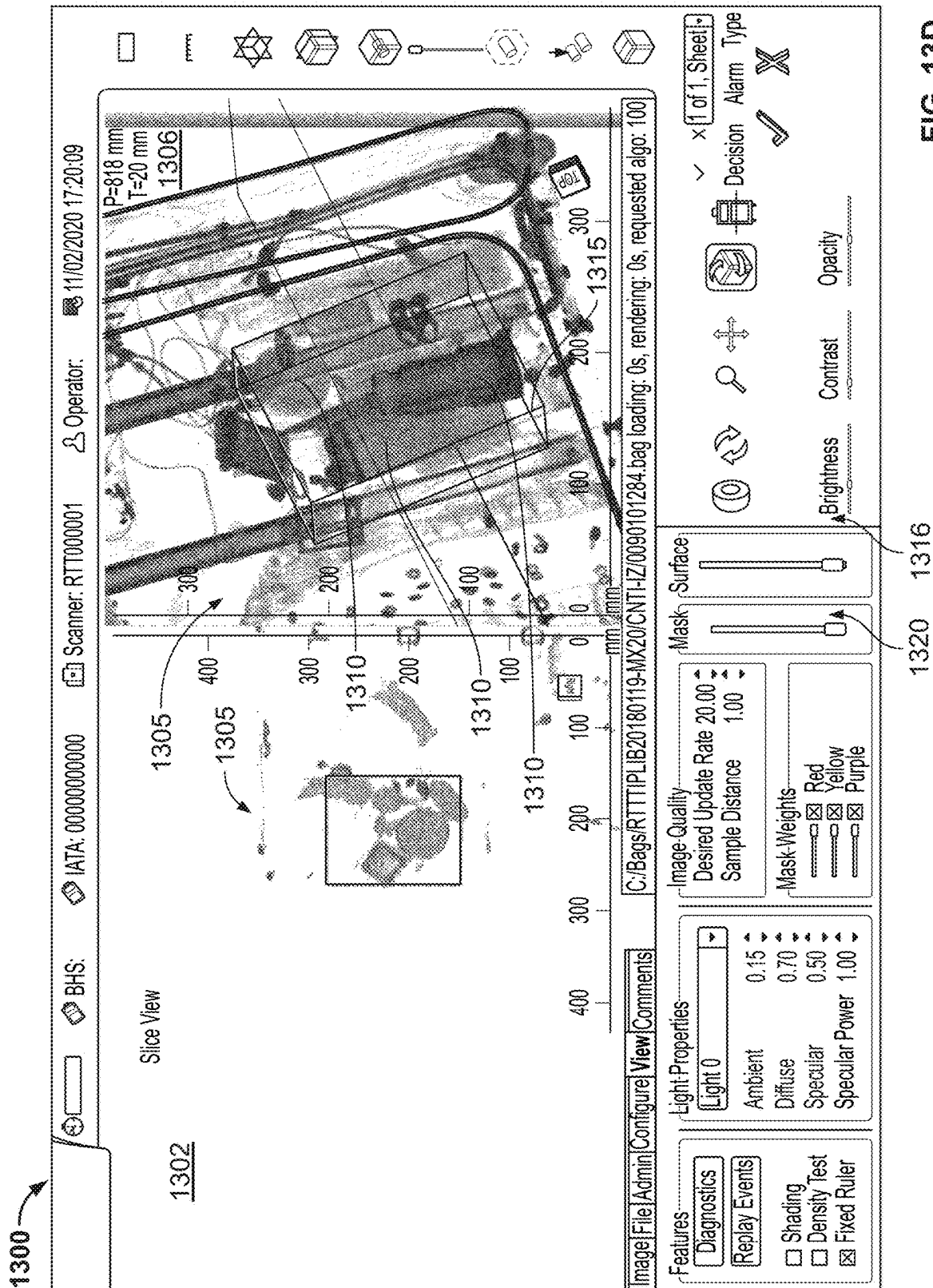


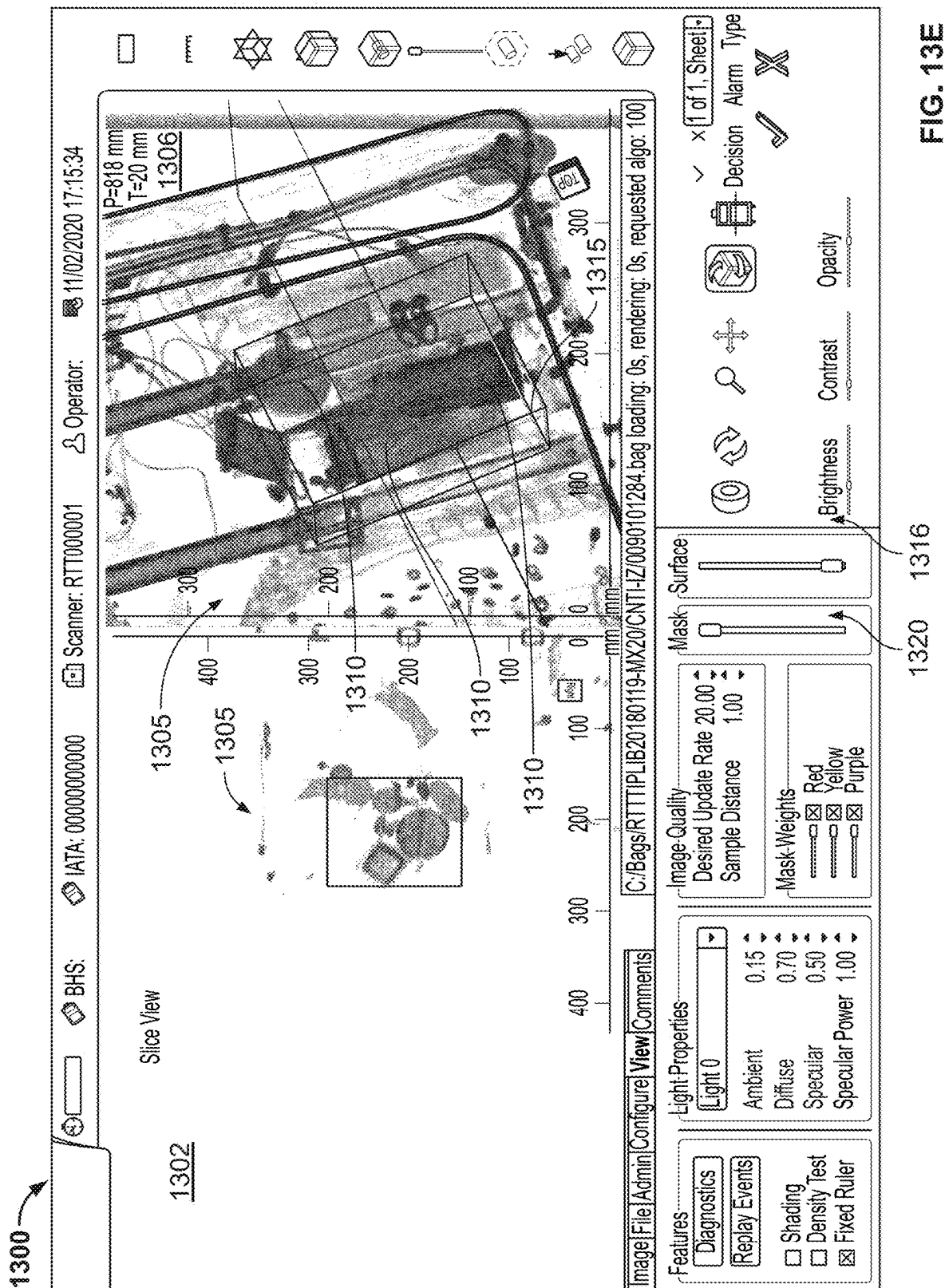
FIG. 12











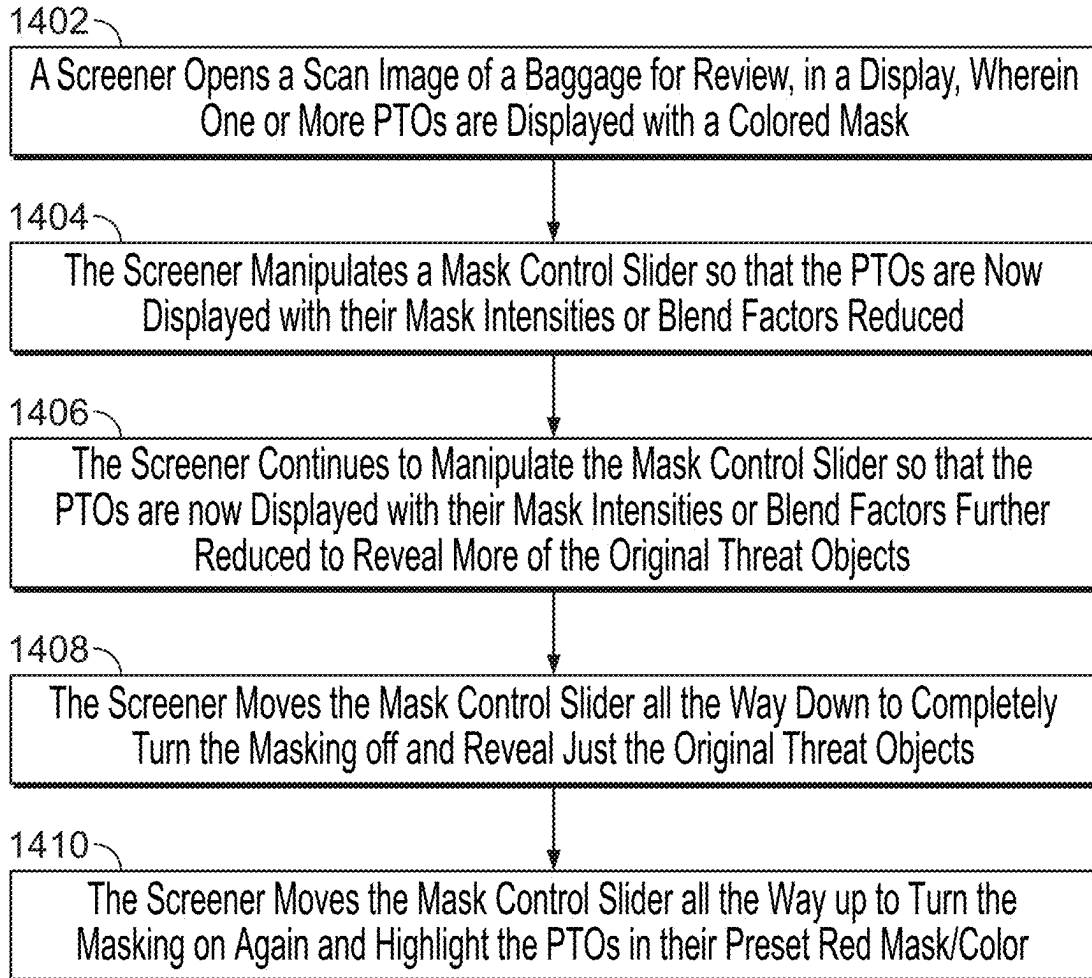


FIG. 14

**METHODS AND SYSTEMS FOR
GENERATING THREE-DIMENSIONAL
IMAGES THAT ENABLE IMPROVED
VISUALIZATION AND INTERACTION WITH
OBJECTS IN THE THREE-DIMENSIONAL
IMAGES**

CROSS-REFERENCE

The present application is a division application of U.S. patent application Ser. No. 18/156,159, titled “Methods and Systems for Generating Three-Dimensional Images that Enable Improved Visualization and Interaction with Objects in the Three-Dimensional Images” and filed on Jan. 18, 2023, which is a division application of U.S. patent application Ser. No. 16/928,983, of the same title, filed on Jul. 14, 2020, and issued as U.S. Pat. No. 11,594,001 on Feb. 28, 2023, which relies on, for priority, United States Patent Provisional Application No. 62/963,494, titled “Methods and Systems for Generating Three-Dimensional Images that Enable the Improved Visualization and Interaction with Objects in the Three-Dimensional Image” and filed on Jan. 20, 2020, all of which are herein incorporated by reference in their entirety.

FIELD

The present specification relates to X-ray scanning systems. More particularly, the present specification relates to X-ray computed tomography inspection systems having improved graphical user interface displays of three-dimensional images that enable improved visualization and interaction with objects in the three-dimensional images.

BACKGROUND

X-ray computed tomography (CT) scanners have been used in security screening in airports for several years. A conventional system comprises an X-ray tube that is rotated about an axis with an arcuate X-ray detector which is also rotated, at the same speed, around the same axis. The conveyor belt on which the baggage is carried is placed within a suitable aperture around the central axis of rotation and moved along the axis as the tube is rotated. A fan beam of X-radiation passes from the source through the object to be inspected and subsequently to the X-ray detector array.

The X-ray detector array records the intensity of X-rays passed through the object to be inspected at several locations along its length. One set of projection data is recorded at each of a number of source angles. From these recorded X-ray intensities, it is possible to form a tomographic (cross-sectional) image, typically by means of a filtered back projection algorithm. In order to produce an accurate tomographic image of an object, such as a bag or package, there is a requirement that the X-ray source pass through every plane through the object. In the arrangement described above, this is achieved by the rotational scanning of the X-ray source, and the longitudinal motion of the conveyor on which the object is carried.

In this type of system the rate at which X-ray tomographic scans can be collected is dependent on the speed of rotation of the gantry that holds the X-ray source and detector array. In a modern CT gantry, the entire tube-detector assembly and gantry will complete two to four revolutions per second. This allows up to four or eight tomographic scans to be collected per second, respectively.

As the state-of-the-art has developed, the single ring of X-ray detectors has been replaced by multiple rings of detectors. This allows many slices (typically 8) to be scanned simultaneously and reconstructed using filtered back projection methods adapted from the single scan machines. With a continuous movement of the conveyor through the imaging system, the source describes a helical scanning motion about the object. This allows a more sophisticated cone-beam image reconstruction method to be applied that can in principle offer a more accurate volume image reconstruction.

Some conventional CT scanners comprise non-rotating stationary gantry systems, which project X-ray beams from fixed, stationary sources at the subjects to be scanned. These systems include one or more spatially distributed X-ray sources for emitting X-rays and one or more X-ray detectors for detecting the X-rays. Multiple X-ray sources are required to be activated at the same time to produce a fan beam of X-rays in order to create a three-dimensional scanned image of an object. Stationary gantry systems may use anywhere from a dozen to a few hundred X-ray sources to produce a scanned image that varies in quality depending on the number of X-ray sources used. Non-rotating gantry CT scanners are also used in medical imaging applications to capture detailed three-dimensional (3D) images of subjects, at high speeds.

Real-time Tomography (RTT) is a new generation of X-ray systems that implement multi-emitter X-ray sources with more than one cathode or electron gun and one or more high voltage anodes within a single vacuum tube, envelope or X-ray tube. In this system, a multi-emitter X-ray source allows non-sequential motion of an X-ray beam about an object under inspection through the use of multiple grid controlled cathodes which can be excited in any chosen sequence, the electron beam from each source being directed to irradiate anode sections which are distributed around the object under inspection. This allows non-helical source trajectories to be constructed at high speeds consistent with the requirements for dynamic and high-throughput object imaging. Additionally, the rapid switching of cathodes under electrostatic control enables a fast movement of the effective focal spot of the X-ray tube and rapid generation of sets of tomographic X-ray scan data without the use of moving parts.

Human visual system has a capability of viewing across a great bandwidth for interpretation of a scene. 3D datasets presented by the current scanning and imaging systems pose a challenge for visualization that has not yet been overcome. The amount of 3D data to be shown all at once in a single view is limited. Objects of interest are often hidden within the data and occluded. Visualization of CT data of a large bag in airport screening contains one or more regions of interest (ROI) that may be potential threat objects (PTOs), occluded by surrounding clutter in a baggage that is being scanned. These ROIs often cannot be discriminated from occluding regions. Simply extracting the ROIs or PTOs from the image may not be helpful because one typically loses context upon isolating the ROI or PTOs. Visualization of small ROIs without surrounding information is often meaningless. Surrounding context is particularly important when visualizing an object within a 3D scene because humans comprehend objects and spatial relationships between objects based on depth cues. In order for the user to correctly interpret the PTO (the focus), interact with it or orient oneself, the user simultaneously needs a detailed depiction (focus) along with a general overview (context).

While CT X-ray scanners are able to produce 3D images of objects under inspection and/or of medical subjects, they are limited in their ability to process complex 3D images and portions thereof, for a relatively more focused and contextual analysis—to resolve security threats and, of the regions of interest in medical applications. Therefore, there is a need for improving efficiency and usability of the analysis of threat objects and/or anatomical structures embedded in a complex 3D image.

SUMMARY

The following embodiments and aspects thereof are described and illustrated in conjunction with systems, tools and methods, which are meant to be exemplary and illustrative, and not limiting in scope. The present application discloses numerous embodiments.

In some embodiments, the present specification discloses a method of scanning an object using an X-ray scanner having a scanning volume, comprising: transporting the object through the scanning volume using a conveyor; irradiating the object with X-rays generated by a plurality of X-ray sources arranged around the scanning volume; detecting X-rays transmitted through the object using a detector array positioned between the X-ray source and the scanning volume, wherein the detector array has a plurality of multi-energy detector modules arranged around the scanning volume; analyzing reconstructed image data of the object being inspected to identify a potential threat object within the object by classifying the potential threat object, wherein the classifying comprises identifying voxels of the potential threat object in the voxels of the image of the object; and displaying the reconstructed image data as an image on a display, wherein the displaying comprises isolating the potential threat object with spatial and contextual information relative to the object.

Optionally, the classifying comprises visually highlighting the voxels of the potential threat object within a visually perceptible bounding box in the image.

Optionally, the classifying comprises visually highlighting the voxels of the potential threat object with optical properties that are different from optical properties of remaining voxels of the object in the image.

Optionally, the optical properties comprise illuminating the potential threat object with a color that is different from the remaining voxels of the object in the image, edge highlighting, or sampling.

Optionally, the highlighting with optical properties comprises illuminating the potential threat object with an opacity that is different from the remaining voxels of the object in the image.

Optionally, the plurality of X-ray sources are contained in a multi-focus X-ray source arranged around the scanning volume.

Optionally, the plurality of X-ray sources are a plurality of X-ray source points in the multi-focus X-ray source.

Optionally, the analyzing comprises animating the image of the object and the potential threat object.

Optionally, the isolating the potential threat object comprises magnifying the image of the potential threat object.

Optionally, the isolating comprises removing portions of the reconstructed image data that occlude a view of the potential threat object by, at least one of, executing a view dependent virtual cut-away or rendering occluding portions of the reconstructed image data as transparent.

Optionally, the method further comprises displaying the image of the potential threat object in a foreground by

moving a visual position of the potential threat object from a first position on the display on to a second position on the display, wherein the second position appears closer to a viewer on the display relative to the first position.

Optionally, isolating the potential threat object comprises diminishing a remainder of the image after visually highlighting the potential threat object.

Optionally, the method further comprises displaying a portion of the image that is not the potential threat object in a background by moving a visual position of the portion of the image that is not the potential threat object on the display on to a second position on the display, wherein the second position appears farther away to a viewer on the display relative to the first position.

Optionally, the method further comprises displaying the portion of the image that is not the potential threat object in a thumbnail on the display.

Optionally, the method further comprises displaying the portion of the image that is not the potential threat object in a corner of a display.

Optionally, the analyzing comprises enabling a user to interact with the reconstructed image data for at least one of classifying and isolating the image of the potential threat object with spatial and contextual information relative to the image of the object.

Optionally, the analyzing comprises enabling a user to interact with the reconstructed image data by receiving a first physical touch or clicking on the mouse of an area of a display that depicts a portion of the reconstructed image data and, in response to the first physical touch, visually isolating said portion of the reconstructed image data from a remainder of the portion of the reconstructed image data.

Optionally, the visual isolating comprises replacing the portion of the reconstructed image data with a transparent void having dimensions equivalent to the portion of the reconstructed image data.

Optionally, the visual isolating comprises modifying a visual scale of the portion of the reconstructed image data to make the visual presentation of the portion of the reconstructed image data visually larger than the portion of the reconstructed image data prior to the first physical touch.

Optionally, a plurality of guide lines maintain spatial and contextual comprehension between said visually isolated portion and said remainder of the reconstructed image data of the object.

Optionally, a scale is provided along the isolated object to indicate the physical measurement of the object

Optionally, in response to a second physical touch of the area of the display that depicts the portion of the reconstructed image data, the method further comprises visually placing the portion of the reconstructed image data back into the thumbnail such that it is in the same visual configuration as shown prior to the first physical touch.

Optionally, the method further comprises enabling the user to interactively define any portion of the visually displayed reconstructed image data and select any portion of the visually displayed reconstructed image data to be isolated such that the potential threat object is in the foreground and a remainder of the image is visually positioned in a corner of the display as a thumbnail.

Optionally, the display comprises a touch screen and the interactive identification is enabled by touch operations on the touch screen.

In some embodiments, the present specification discloses a method of enabling at least first and second operators to consecutively associate audio comments with a scan image of an object, the method comprising: viewing, by said first

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operator, said scan image on a graphical interface, wherein said graphical interface includes an audio button; activating, by said first operator, said audio button to begin recording audio comments with reference to said scan image while simultaneously reviewing and maneuvering said scan image in said graphical interface; deactivating, by said first operator, said audio button to generate a first audio file of said audio comments of said first operator, save said first audio file in association with said scan image and display a first icon on said graphical interface, wherein said first icon is indicative of said first audio file, wherein said first audio file is associated with an identification of said first operator, and wherein said first audio file has an associated date and time of recording by said first operator; viewing, by said second operator, said scan image on said graphical interface, wherein said graphical interface includes said audio button and said first icon; activating, by said second operator, said first icon to enable said second operator to listen to said first audio file while simultaneously reviewing and maneuvering said scan image in said graphical interface; activating, by said second operator, said audio button to begin recording audio comments with reference to said scan image while simultaneously continuing to review and maneuver said scan image in said graphical interface; and deactivating, by said second operator, said audio button to generate a second audio file of said audio comments of said second operator, save said second audio file in association with said scan image and display a second icon on said graphical interface, wherein said second icon is indicative of said second audio file, wherein said second audio file is associated with an identification of said second operator, and wherein said second audio file has an associated date and time of recording by said second operator.

In some embodiments, the present specification discloses a method of reviewing a scan image of an object, the method comprising: presenting, to an operator, said scan image on a graphical interface, wherein a plurality of voxels in said scan image are highlighted in at least one mask, said plurality of voxels being indicative of at least one potential threat item, wherein said at least one mask has a first intensity of color of said mask, and wherein said graphical interface has an actuator to enable modulation of said first intensity to a plurality of intensities between a maximum and a minimum intensity; manipulating, by said operator, said actuator to modulate said first intensity to a second intensity of the color of said mask, wherein said second intensity is less than said first intensity; further manipulating, by said operator, said actuator to modulate said second intensity to a third intensity of the color of said mask, wherein said third intensity is less than said second intensity; and further manipulating, by said operator, said actuator to modulate said third intensity to said first intensity of the color of said mask.

Optionally, said first intensity is representative of a least level of transparency of the color of said mask. Optionally, said third intensity is representative of a highest level of transparency of the color of said mask. Optionally, at said second intensity, a visibility of said at least one potential threat item increases compared to a visibility of said at least one potential threat item at said first intensity.

The aforementioned and other embodiments of the present specification shall be described in greater depth in the drawings and detailed description provided below.

BRIEF DESCRIPTION OF THE DRAWINGS

These and other features and advantages of the present invention will be further appreciated, as they become better

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understood by reference to the detailed description when considered in connection with the accompanying drawings:

FIG. 1 is a longitudinal schematic view of a real time tomography security scanning system having a circular locus of source points, produced by conventional systems;

FIG. 2A is a perspective view of a scanning unit, in accordance with an embodiment of the present specification;

FIG. 2B is a cross-sectional view of a housing/enclosure of the scanning unit of FIG. 2A, comprising a plurality of X-ray sources points and detectors, arranged in a substantially rectangular shape around a scanning volume, in accordance with an embodiment of the present specification;

FIG. 3 illustrates an exemplary image of a scanned object and a potential threat object (PTO) within the object, in accordance with some embodiments of the present specification;

FIG. 4A illustrates an exemplary step in the process of isolating image of a PTO in a 3D view within a display, in accordance with some embodiments of the present specification;

FIG. 4B illustrates an exemplary step of a user selecting (such as by clicking or touching) focus object to initiate the viewing of PTO in isolation, on a touch screen display, in accordance with some embodiments of the present specification;

FIG. 5A illustrates a first step in an exemplary sequence of steps that provide an isolated display of a PTO in a 3D display, while retaining spatial and contextual information of the PTO relative to a baggage, in accordance with some embodiments of the present specification;

FIG. 5B illustrates a second step in an exemplary sequence of steps that provide an isolated display of a PTO in a 3D display, while retaining spatial and contextual information of the PTO relative to a baggage, in accordance with some embodiments of the present specification;

FIG. 5C illustrates a third step in an exemplary sequence of steps that provide an isolated display of a PTO in a 3D display, while retaining spatial and contextual information of the PTO relative to a baggage, in accordance with some embodiments of the present specification;

FIG. 5D illustrates a fourth step in an exemplary sequence of steps that provide an isolated display of a PTO in a 3D display, while retaining spatial and contextual information of the PTO relative to a baggage, in accordance with some embodiments of the present specification;

FIG. 5E illustrates a fifth step in an exemplary sequence of steps that provide an isolated display of a PTO in a 3D display, while retaining spatial and contextual information of the PTO relative to a baggage, in accordance with some embodiments of the present specification;

FIG. 5F illustrates a sixth step in an exemplary sequence of steps that provide an isolated display of a PTO in a 3D display, while retaining spatial and contextual information of the PTO relative to a baggage, in accordance with some embodiments of the present specification;

FIG. 6 illustrates a flow chart describing the process of FIGS. 5A to 5F, in accordance with some embodiments of the present specification;

FIG. 7 illustrates the optional use of guide lines from an outlining box of a PTO image in focus in a 3D view, to its extracted region of interest in the context of the baggage image, to facilitate correct spatial comprehension during interaction with the isolated PTO, in accordance with some embodiments of the present specification;

FIG. 8A illustrates an exemplary initial 3D view of a transitional animation that displays the transition of an

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isolated PTO back in to its region of interest within a baggage, in accordance with some embodiments of the present specification;

FIG. 8B illustrates an exemplary 3D view with the final magnified image of the baggage including the PTO, in accordance with some embodiments of the present specification;

FIG. 9A illustrates a focus including a PTO within a region of interest bound by a box inside a context of a baggage, in accordance with some embodiments of the present specification;

FIG. 9B illustrates edge highlighting of the objects within the region of interest in the box, in accordance with some embodiments of the present specification;

FIG. 9C illustrates voxels within the box that includes the focus, shaded and rendered with different optical properties, lambertian shading, ambience, and with edge highlighting, in accordance with some embodiments of the present specification;

FIG. 9D illustrates the PTO (focus) displayed as unoccluded or unhindered, along with contextual information, whereby everything in front of the focus is dynamically cut away or removed, thus creating a virtual vision channel to visualize deep-seated PTOs;

FIG. 9E is another illustration of the (PTO) focus displayed as un-occluded or unhindered, along with contextual information, whereby everything in front of the focus is dynamically cut away or removed, thus creating a virtual vision channel to visualize deep-seated PTOs;

FIG. 9F is a first illustration wherein a user begins defining a region of interest around an object such that it can be isolated in the same manner as a PTO;

FIG. 9G is a second illustration wherein the region of interest is expanded;

FIG. 9H is a third illustration wherein the region of interest is further defined and moved;

FIG. 10 illustrates a flow chart for an exemplary process adapted by some embodiments of the present specification to scan an object;

FIG. 11A illustrates a first step in a method for enabling at least two security screeners to record audio comments related to a scan image of a bag, in accordance with some embodiments of the present specification;

FIG. 11B illustrates a second step in a method for enabling the at least two security screeners to record audio comments related to the scan image of the bag, in accordance with some embodiments of the present specification;

FIG. 11C illustrates a third step in a method for enabling the at least two security screeners to record audio comments related to the scan image of the bag, in accordance with some embodiments of the present specification;

FIG. 11D illustrates a fourth step in a method for enabling the at least two security screeners to record audio comments related to the scan image of the bag, in accordance with some embodiments of the present specification;

FIG. 11E illustrates a fifth step in a method for enabling the at least two security screeners to record audio comments related to the scan image of the bag, in accordance with some embodiments of the present specification;

FIG. 11F illustrates a sixth step in a method for enabling the at least two security screeners to record audio comments related to the scan image of the bag, in accordance with some embodiments of the present specification;

FIG. 12 is a flowchart describing the process of FIGS. 11A through 11F, in accordance with some embodiments of the present specification;

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FIG. 13A illustrates a first step in a method for enabling a security screener to modulate the mask intensity of highlighted one or more PTOs in a scan image of a bag, in accordance with some embodiments of the present specification;

FIG. 13B illustrates a second step in a method for enabling the security screener to modulate the mask intensity of the highlighted one or more PTOs in the scan image of the bag, in accordance with some embodiments of the present specification;

FIG. 13C illustrates a third step in a method for enabling the security screener to modulate the mask intensity of the highlighted one or more PTOs in the scan image of the bag, in accordance with some embodiments of the present specification;

FIG. 13D illustrates a fourth step in a method for enabling the security screener to modulate the mask intensity of the highlighted one or more PTOs in the scan image of the bag, in accordance with some embodiments of the present specification;

FIG. 13E illustrates a fifth step in a method for enabling the security screener to modulate the mask intensity of the highlighted one or more PTOs in the scan image of the bag, in accordance with some embodiments of the present specification; and

FIG. 14 is a flowchart describing the process of FIGS. 13A through 13E, in accordance with some embodiments of the present specification.

DETAILED DESCRIPTION

Embodiments of the present specification provide methods and systems for efficiently visualizing and interacting with, and therefore isolating Potential Threat Objects (hereinafter, referred to as a "PTO") within baggage along with its surrounding spatial and contextual information. An imaging system, such as a stationary CT scanner or a rotating CT scanner, is provided with an object, such as baggage, or a medical subject and generates a three-dimensional image defined by a plurality of voxels. In embodiments, semantic information, such as including and not limited to semantic classification and a bounding box, of a PTO within the scan of the baggage or of a region of interest within the scan of an anatomical structure of the medical subject are identified and provided to the system. A context of the PTO or the region of interest is provided by volume-rendering the image of the whole baggage or the anatomical structure, while the PTO or the region of interest is displayed in a focus region.

In embodiments, the system enables a user to interactively select a PTO or a region of interest from the scan, by clicking on it from the display or otherwise interacting with the scan to select the image of the PTO or the region of interest. In alternative methods of interaction, the user may touch the image of the PTO or the region of interest on a touch screen, so as to select the image of the PTO or the region of interest. It should be noted herein that any object may be extracted and manipulated from the scan, and that the selection is not limited to the image of the PTO or region of interest. In further alternative embodiments for methods of interaction, a user may employ methods of eye-tracking and dwell. In still further alternative embodiments, a user may employ various augmented reality techniques, including "pull and grab". In embodiments, the user's selection brings the PTO (or the region of interest) and/or the focus region into foreground, while the whole baggage (or the anatomical structure) providing a spatial and contextual background for the PTO (or the region of interest), fades into

a corner of the display. The user may further interact with the selected image of the PTO or the region of interest including by rotating the image to view it from different angles and perspectives. In embodiments, rotating the selected PTO image or region of interest image simultaneously rotates contextual image of the whole baggage. In some embodiments, the contextual background is rendered with the image of the selected PTO or the extracted region of interest, so as to indicate that the PTO or the region of interest has been brought into the foreground.

In some embodiments, the system and method of the present specification highlight one or more focus regions by rendering them with different optical properties and different light transport parameters such as including and not limited to a different transfer function, increased ray sampling density during ray casting, and accentuated edge highlighting with gradient opacity transfer functions. Additionally, within the PTO bounding box, voxels that are classified as threats are highlighted with different optical properties, such that it stands out from the rendering of the contextual background.

The present specification is directed towards multiple embodiments. The following disclosure is provided in order to enable a person having ordinary skill in the art to practice the invention. Language used in this specification should not be interpreted as a general disavowal of any one specific embodiment or used to limit the claims beyond the meaning of the terms used therein. The general principles defined herein may be applied to other embodiments and applications without departing from the spirit and scope of the invention. Also, the terminology and phraseology used is for the purpose of describing exemplary embodiments and should not be considered limiting. Thus, the present invention is to be accorded the widest scope encompassing numerous alternatives, modifications and equivalents consistent with the principles and features disclosed. For purpose of clarity, details relating to technical material that is known in the technical fields related to the invention have not been described in detail so as not to unnecessarily obscure the present invention.

In the description and claims of the application, each of the words “comprise” “include” and “have”, and forms thereof, are not necessarily limited to members in a list with which the words may be associated. It should be noted herein that any feature or component described in association with a specific embodiment may be used and implemented with any other embodiment unless clearly indicated otherwise.

It should be noted herein that while the embodiments have been explained with reference to a non-rotating or stationary gantry CT scanner, the methods of the present specification can also be used in rotating gantry CT scanners. While various embodiments have been explained with reference to identifying a PTO (Potential Threat Object), it should be appreciated that the embodiments are not limited to threat objects and that the embodiments of the present specification are equally applicable to identifying regions/areas of interest in anatomical structures in medical subjects for diagnostic purposes as well as in conjunction with industrial applications.

It should be further noted herein that while some embodiments are explained with reference to a particular CT scanner system having a rectangular tunnel geometry, the methods of the present specification are not limited to such embodiments. Some embodiments of the present specification are explained with reference to a CT scanner and persons skilled in the art will appreciate that the various

embodiments can be used in different types of scanners, provided they generate three dimensional images defined by a plurality of voxels.

In embodiments of the specification, the term “extracted” is defined as visually isolating or separating a region from the remainder of the image. In an embodiment, for example, extraction is accompanied by a modification of the remainder of the image to visually compensate for the absence of the extracted region by a) shrinking the area (in the remainder of the image) from which the extracted region was removed, b) coloring the area (in the remainder of the image) a different color relative to the image before extraction, c) blacking out the area (in the remainder of the image) relative to the image before extraction, d) whitening out the area (in the remainder of the image) relative to the image before extraction or e) overlaying a pattern (such as cross hatch) in the area (in the remainder of the image) relative to the image before extraction.

FIG. 1 illustrates a conventional inspection system having a circular locus of source points. Referring to FIG. 1, a concourse baggage scanning system 6 comprises a scanning unit 8 which includes a multi-focus X-ray source 10 and X-ray detector array 12. The source 10 comprises a large number of source points 14 positioned in respective, spaced locations on the source, and arranged in a full 360 degree circular array about the X-X axis of the system (which is parallel to the conveyor belt 20). It will be appreciated that curved arrays covering less than the full 360 degree angle can also be used. The source 10 can be controlled to produce X-rays from each of the source points 14 in each of the source units individually whereby X-rays from each source point 14 are directed inwards through the scanning region 16 within the circular source 10. The source 10 is controlled by a control unit 18 which controls the applied electrical potentials (to the grid wires) and hence controls the emission of X-rays from each of the source points 14.

The multi-focus X-ray source 10 allows the electronic control circuit 18 to be used to select which of the many individual X-ray source points 14 within the multi-focus X-ray source is active at any moment in time. Hence, by electronically scanning the multi-focus X-ray tube, X-ray source virtual “motion” is created with no actual physical movement of mechanical parts. In this case, the angular velocity of source rotation can be increased to levels that simply cannot be achieved when using conventional rotating X-ray tube assemblies. This rapid rotational scanning translates into an equivalently speeded up data acquisition process and, as a result, fast image reconstruction.

The detector array 12 is also circular and arranged around the axis X-X in a position that is slightly offset in the axial direction from the source 10. The source 10 is arranged to direct the X-rays it produces through the scanning region 16 towards the detector array 12 on the opposite side of the scanning region. The paths 17 of the X-ray beams therefore pass through the scanning region 16 in a direction that is substantially, or almost, perpendicular to the scanner axis X-X, crossing each other near to the axis. The volume of the scanning region that is scanned and imaged is therefore in the form of a thin slice perpendicular to the scanner axis X-X. The source is scanned so that each source point emits X-rays for a respective period, the emitting periods being arranged in a predetermined order. As each source point 14 emits X-rays, the signals from the detectors 12, which are dependent on the intensity of the X-rays incident on the detector, are produced, and the intensity data that the signals provide are recorded in a memory. When the source has

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completed its scan the detector signals can be processed to form an image of the scanned volume.

A conveyor belt **20** moves through the imaging volume, from left to right, as seen in FIG. 1, parallel to the axis X-X of the scanner. X-ray scatter shields **22** are located around the conveyor belt **20** upstream and downstream of the main X-ray system to prevent operator dose due to scattered X-rays. The X-ray scatter shields **22** include lead rubber strip curtains **24** at the open ends of the system such that the item **26** under inspection is conveyed through one curtain on entering the inspection region and another curtain upon leaving the inspection region. In the integrated system shown, the main electronic control system **18**, a processing system **30**, a power supply **32** and cooling racks **34** are shown mounted underneath the conveyor **20**. The conveyor **20** is arranged to be operated normally with a continuous scanning movement at constant conveyor speed, and typically has a carbon-fiber frame assembly within the imaging volume.

It should be noted that the systems described throughout this specification comprise at least one processor (such as processing system **30**) to control the operation of the system and its components. It should further be appreciated that the at least one processor is capable of processing programmatic instructions, has a memory capable of storing programmatic instructions, and employs software comprised of a plurality of programmatic instructions for performing the processes described herein. In one embodiment, the at least one processor is a computing device capable of receiving, executing, and transmitting a plurality of programmatic instructions stored on a volatile or non-volatile computer readable medium.

In accordance with an embodiment of the present specification, FIG. 2A is a perspective view of a scanning unit **200**, shown from a first side **245**, comprising a substantially rectangular housing/enclosure **201** for housing a plurality of X-ray source points and detectors. It should be appreciated that, in alternate embodiments, the housing **201** may have a quadrilateral shape, such as, but not limited to, a square. An object under inspection is conveyed through a first open end or scanning aperture **203**, enters an inspection region **206**, and exits through a second open end (opposite to the first open end **203**). In accordance with an embodiment, both feed and return conveyor loops pass through a space **216** just below the inspection region **206**, while space or compartment **240** is reserved in the base of the scanning system (approximately 200 mm deep) to accommodate automated return of trays when integrated with an automatic tray return handling system. The scanning unit **200** has an external body comprising the components stated above within said body. In embodiments, the body of unit **200** is shaped similar to a large elongated right rectangular prism, or a rectangular cuboid with curved corners. In some embodiments, the unit **200** is an extension of the shape of housing/enclosure **201**. In embodiments, the inspection region **206** positioned within housing **201** is shaped similar to housing **201**. In some embodiments, a narrow projection **290** encompasses three external surfaces of the unit **200**.

FIG. 2B illustrates a cross-sectional view of the housing **201** of a scanning unit **200**, comprising a plurality of X-ray source points and detectors arranged in a substantially rectangular shape around a scanning volume, in accordance with first and second embodiments of the present specification. In various embodiments, the rectangular housing **201** has width ranging from 800 mm to 1400 mm and a height ranging from 600 mm to 1500 mm. In various embodiments, the housing **201** is configured to define an imaging volume

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or inspection tunnel **206**, which is also rectangular, that has a width ranging from 500 mm to 1050 mm and a height ranging from 300 mm to 1050 mm. It should be appreciated that, in alternate embodiments, the plurality of X-ray source points and detectors can be arranged in other quadrilateral shapes, such as, but not limited to, a square shape. It should be appreciated that the rectangular, quadrilateral, or square shape may also have rounded edges and encompasses shapes known as rounded rectangles, squircles, or rectellipses.

The scanning unit **200** comprises a multi-focus X-ray source **202** and X-ray detector array **204** enclosed within housing **201**. The source **202** comprises a large number of source points (or, in an embodiment, electron guns **220**) in locations spaced about the source **202**, and arranged in a substantially non-circular, such as rectangular, geometry around an imaging or inspection volume **206**, in accordance with an embodiment. In embodiments, the X-ray detector array **204** is positioned between the X-ray source points **220** and the imaging volume **206** such that the source points **220** and the detector array **204** surround the imaging volume **206**.

A conveyor belt **208** carries objects/luggage to be inspected through the imaging volume **206** along a longitudinal axis of the scanning unit **200**. In an embodiment, the conveyor belt **208** has a speed of 0.5 m/s which is about twice the speed of conventional X-ray systems that typically operate at a speed of about 0.25 m/s and is about three times the speed of conventional rotating gantry systems that typically operate at a speed of about 0.15 m/s. In various embodiments, the conveyor belt **208** has a speed ranging from 0.1 m/s to 1.0 m/s. Both feed and return conveyor loops pass through the base **216** of the imaging volume **206**, having a depth of approximately 50 mm while space **240** (approximately 200 mm deep and having a width equal to that of the base **216** of the imaging volume **206**) is reserved in the base of the scanning unit **200**, to accommodate automated return of trays when integrated with an automatic tray return handling system, in accordance with some embodiments. The conveyor and feed return loops both pass through base **216** of imaging volume **206**. In contrast, trays that have been conveyed through the inspection or imaging volume **206** by the conveyor **208** are returned back through region **240**, which ranges from 100 mm to 300 mm deep and is preferably 200 mm deep.

In various embodiments, the rectangular housing **201** has width ranging from 800 mm to 1400 mm and a height ranging from 600 mm to 1500 mm. In embodiments, the housing **201** has a maximum width of 920 mm and a maximum height of 720 mm. In various embodiments, the housing **201** is configured to define an imaging volume or inspection tunnel **206**, which is also rectangular, that has a width ranging from 500 mm to 1050 mm and a height ranging from 300 mm to 1050 mm. In some embodiments, the housing **201** is configured to define an imaging volume or inspection tunnel **206** that is approximately 620 mm in width and approximately 420 mm in height.

In an embodiment, as shown in FIG. 2B, X-ray source **202** comprises 256 electron guns **220**, grouped in units of 16, substantially equidistantly spaced around the imaging volume **206** on a 12 mm pitch (that is, a center-to-center spacing between adjacent electron guns is 12 mm). In various embodiments, the X-ray source **202** comprises 64 to 2048 electron guns grouped in 4 to 32 units of electron guns. In various embodiments, the electron guns **220** are spaced on a pitch ranging from 10 mm to 14 mm. In this configuration, every emission source point has a different field of view (FOV). In various embodiments, the X-ray sources emit fan

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beams which have different beam angles based on the location of the X-ray source points with respect to the imaging volume.

As shown in FIG. 2B, a plurality of support means **214a**, **214b**, and **214c**, positioned at points along the periphery of the imaging volume **206**, are provided for supporting the X-ray source **202**. In an embodiment, the support means **214b** and **214c** are also used to provide coolant and power to the X-ray source **202** and the scanning system **200**, respectively.

While FIGS. 1, 2A, and 2B describe non-rotating CT scanning systems, embodiments of the present specification are applicable to rotating CT scanners and other types of X-ray scanning and imaging systems.

FIG. 3 illustrates an exemplary image of a scanned object and a potential threat object within the object, in accordance with some embodiments of the present specification. Embodiments of the present specification generate a scanned image of an object, such as a baggage, which is displayed on a display with accompanying semantic information. The semantic information can include a classification of the image voxels in the object in different regions, such as including and not limited to one or more regions of interest and a background. In addition, the voxels in the image may be described as an “obstruction”, wherein in embodiments, an “obstruction” is defined as a region where X-rays do not penetrate to provide sufficient information such that a classification can be made (thus, an “undeterminable” region), as determined by the algorithm. A region of interest may include a potential threat object (PTO), which may be positioned behind an obstructing object in the image. The background comprises spatial and contextual information relative to the region of interest. Referring to FIG. 3, a display **300** is used to display scanned images of the object. In some embodiments, the display includes a slice view **302** of the scanned object **304**, shown on the left side of the display **300**; and a 3D view **306** shown on the right side. The exemplary object shown in FIG. 3 is a baggage including two regions of interest highlighted within colored bounding boxes **308** and **310**. In some embodiments, the bounding boxes for different semantic classes are color coded. In one embodiment, bounding boxes that are displayed around regions of interest are shown in yellow color. Each region of interest **308**, **310** includes a PTO **312**, **314**, respectively, which are highlighted in a color. In some embodiments, the PTOs **312**, **314** are highlighted in red color. In some embodiments, metallic objects are highlighted in blue color.

The display **300** includes a menu **316** with buttons or icons that may be used by a user to interface with the images shown in the display **300**. The buttons in the menu **316** are interactive and used to modify or edit the displayed images by selecting the button. In some embodiments, the user may select a button or any item from the menu **316** by hovering a pointer over the option and clicking the option. Alternatively, the user may select an option by touching the option when the display is provided on a touchscreen. In some alternative embodiments, voice commands are provided to interface with the images through the menu **316**. It should be noted that, in embodiments, voice command may be used to record and playback comments on the images.

In embodiments, the display **300** presents at least one scale **320** (**320x**, **320y**) in association with the identification of the regions of interest **308**, **310** that respectively include PTOs **312**, **314**. In other words, once a region of interest or PTO is identified, at least one scale **320** is generated and displayed to allow for easy assessment of a physical measurement of the PTO. In some embodiments, the scale **320**

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includes a horizontal scale **320x** and a vertical scale **320y**. In an embodiment, the horizontal and vertical scales **320x**, **320y** are calibrated or denominated in millimeters in increments of, for example, 200 millimeters. However, the denomination of the scale may be customized in other units such as, for example, centimeters.

It is desirable to view and inspect PTO **312** or **314** more closely and with greater detail. In embodiments, the present specification enables the user to isolate the image of a PTO in the display **300**, so as to examine it in detail. FIG. 4A illustrates an exemplary step in the process of isolating image of a PTO **412** in a 3D view **406** within a display **400**, in accordance with some embodiments of the present specification. The user may hover a mouse cursor **418** over the PTO **412** highlighted within a box **408**, for visual cues. In some embodiments, visual cues are provided in the form of an outline **420** around the PTO **412** with at least one arrow tip **422**, which indicate to the user that the PTO **412** can be brought in to focus. FIG. 4B illustrates an exemplary step of a user clicking or touching a focus object to initiate the viewing of PTO **412** in isolation, on a display (which may, in some embodiments, be a touchscreen display), in accordance with some embodiments of the present specification. In embodiments, any plane on PTO **412** that is primarily located in the baggage, and preferably, with no other object in front of the PTO or occluding the view of the PTO may be used for selecting the PTO.

FIGS. 5A to 5F illustrate an exemplary sequence of steps that provide an isolated display of a PTO **512** in a 3D display **506**, while retaining spatial and contextual information of the PTO **512** relative to a baggage **524**, in accordance with some embodiments of the present specification. FIG. 6 illustrates a flow chart describing the process of FIGS. 5A to 5F, in accordance with some embodiments of the present specification. Referring simultaneously to FIGS. 5A to 5F and FIG. 6, an exemplary sequence of steps that provide an isolated display of PTO **512** in 3D display **506**, while retaining spatial and contextual information of the PTO **512** relative to baggage **524**, are now described. At step **602**, and referring to FIG. 5A, a user clicks on the PTO **512**. In some embodiments, the user selects an option to bring PTO **512** in to focus, such as for example by interfacing with visual cues like an outline **520** around the PTO **512**, also shown in context of FIG. 4B. In embodiments, the action of clicking or touching and thus selecting the PTO **512**, isolates the PTO **512** from the image of the baggage **524** that contains the PTO **512**. In alternative methods of interaction, the user may touch the image of the PTO on a touch screen, so as to select the image of the PTO. In further alternative embodiments for methods of interaction, a user may employ methods of eye-tracking and dwell. In still further alternative embodiments, a user may employ various augmented reality techniques, including “pull and grab”. In embodiments, the object that is isolated is replaced with a void space having a plurality of dimensions, such as length, width, shape, and depth, equivalent to the dimensions of the object, thereby enabling the user to see beyond the object. Thus, objects that may have been otherwise obscured by the PTO are now visible and the context within which they are located is more apparent.

At step **604**, and seen in FIG. 5B, image of the baggage **524** including the PTO **512** is animated into a background within the 3D view **506**. At step **606**, and in FIG. 5C, the image of the baggage **524** including the PTO **512** is animated further into the background within the 3D view **506**, till the image is positioned in the upper right corner of the 3D view **506** in the form of a thumbnail. At step **608**, and in

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FIG. 5D, the isolated image of the PTO 512 is brought forward with animation, in an action that is similar to scooping out the PTO 512 from within the baggage 524. At step 610, and in FIG. 5E, the extracted image of the PTO 512 is magnified and further brought forward towards a foreground within the 3D view 506. During the scooping out and magnification of the PTO 512, the baggage 524 remains positioned as a thumbnail in a corner of the view 506. The thumbnail is a Volume Rendering (VR) of the bag devoid of the PTO 512 voxels. At step 612, and in FIG. 5F, a magnified form of the PTO 512 relative to the baggage 524 is shown. The animation from FIGS. 5A to 5F enables the user to view the PTO in isolation while removing the noise or clutter in proximity of the PTO 512, such as other non-threat objects within baggage 524. FIG. 5F provides a magnified view of the PTO 512 to provide the user with greater details about the PTO 512. The process of scooping out the PTO 512 image from the baggage 524 brings the PTO 512 in to focus, while providing a context of space and orientation and position of the PTO 512 relative to its surrounding objects within the baggage 524. A scaling factor may be adjusted to define a bag image size and PTO size.

In further embodiments, for visualization of the PTO 512 with correct spatial correspondence into the rest of the baggage 524, the contextual thumbnail view's orientation is synchronized with the focus of the PTO 512. Therefore, with the various tools 316 (FIG. 3) available to the user on the display 300, the user may rotate magnified image of the PTO 512 that is in focus, which also correspondingly rotates the thumbnail view of the baggage 524. FIG. 7 illustrates use of guide lines 726 from an outlining box 720 of a PTO 712 image in focus in a 3D view 706, to its extracted region of interest 708 in the context of the baggage 724 image, to facilitate correct spatial comprehension during interaction with the isolated PTO 712, in accordance with some embodiments of the present specification. The guide lines 726 establish a linear connection between the voxels of the PTO 712 and the region of interest within baggage 708 from where the PTO 712 has been extracted. In embodiments, the guide lines 726 portray how an extracted image portion of the PTO 712 is connected to the remaining portion of the image of the baggage 724 from where the PTO 712 was extracted. In some embodiments, the guide lines are visible to the user, whereas in some embodiments they are invisible. In some embodiments, when the user rotates the image of the PTO 712 and changes its orientation, the guide lines 726 guide the corresponding rotation of the baggage 724 in a synchronized fashion, in order to correspondingly change the orientation of the baggage 724. In some embodiments, when the user rotates the image of the PTO 712 and changes its orientation, the baggage 724 remains stationary (that is, does not rotate in synchronization with the rotation of the PTO 712) and instead, the guide lines 726 rotate and twist to show rotational movement of the PTO 712. Therefore, embodiments of the present specification facilitate correct spatial comprehension of a PTO within a scanned baggage, during inspection of the PTO. The guide lines facilitate illustration of how an extracted target object (PTO) is connected or related with the remaining image or surrounding context (baggage) such that when a user isolates and rotates the target object then a) in a first mode the remaining image or surrounding context (baggage) rotates in synchronization with the identified and isolated object (PTO), or b) in a second mode the remaining image or surrounding context (baggage) remains stationary with the guide lines rotating and twisting to show threat object (PTO) movement.

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The user may eventually choose to return to one of the original scanned image and the image of the baggage 724 that includes the image of the PTO 712 within its region 708. FIG. 8A illustrates an exemplary initial 3D view 806 of a transitional animation that displays the transition of an isolated PTO 812 back into its region of interest 808 within a baggage 824, in accordance with some embodiments of the present specification. In some embodiments, the user selects an option from the menu provided along with the 3D view 806 to initiate the transition of the image. In some embodiments, the user clicks/touches the thumbnail of the baggage 824 to initiate the transition. In some embodiments, visual cues, such as and not exclusive to, an arrow 828 is shown within a frame 830 to indicate that the image of the baggage 824 may be brought back into full view within view 806. In some embodiments, the animated sequence shown in FIGS. 5A to 5F are repeated in a reverse order to finally provide a visualization of the baggage 824 including the PTO 812 within its origin region 808. In the animation, the PTO 812 is seen to be pushed back into the baggage 824 at its location and simultaneously the baggage 824 magnifies to fill up the whole 3D view 806. FIG. 8B illustrates an exemplary 3D view 806 with the final magnified image of the baggage 824 including the PTO 812, in accordance with some embodiments of the present specification. Therefore, the user is able to return from a focus view of the PTO 812 to visualize the complete baggage 824 by a single interaction with the display.

Visualization of a focus (PTO) along with surrounding context (baggage) is challenging. In embodiments of the present specification, the focus is brought into prominence using visualization techniques that resort to a distortion of a 'visualization space' such that more space is provided for a certain subset of data comprising the focus. Simultaneously, the rest of the visualization is compressed to still show the rest of the data as a context for improved user orientation. Visualization space traditionally refers to sampling density or increased illumination, or opacity. For instance a background including the context of the baggage (as described in the examples above) can be rendered with an increased transparency to fade it away, while the opacity of the focus including the PTO, which may or may not be in a foreground relative to its context, can be increased to bring it into prominence through the rest of the baggage.

In some embodiments, regions of the image of the context where an object occludes more important structures such as a PTO, can be displayed more sparsely than in those areas where no occlusion occurs. FIG. 9A illustrates a focus including a PTO 912 within a region of interest bound by a box 908 inside a context of a baggage 924, in accordance with some embodiments of the present specification. In the figure, voxels of the baggage 924 (context) that are outside the box 908, are rendered with a reduced opacity relative to the voxels of the objects within the box 908. FIG. 9B illustrates edge highlighting of the objects within the region of interest in the box 908, in accordance with some embodiments of the present specification. FIG. 9C illustrates voxels within the box 908 that includes the focus, shaded and rendered with different optical properties, lambertian shading, ambience, and with edge highlighting, in accordance with some embodiments of the present specification. In embodiments, gradient of dataset in the region of interest bound within the box 908 is computed to multiply the opacity contribution with the normalized gradient magnitude. The computation makes objects within the region of interest stand out by amplifying the transition (edge) between the PTO 912 and surrounding data, resulting in

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edge highlighting. The optical properties of the objects within the box **908** may have different diffuse and specular coefficients relative to the objects outside the box **908**, so as to highlight the region of interest. In some embodiments, a sample distance used for ray integration in a ray caster within the PTO **912** bounding box **908** is increased, resulting in a higher fidelity rendering within the PTO **912**. Therefore, with semantic segmentation less important parts of an image that occlude a significant portion within the image, can be removed to reveal the more important underlying information. Rendering the voxels of PTO **912** with a different colored hue, such as a reddish brown hue, also differentiates it from the context of the baggage **924**. Samplings with several levels of sparsity and different optical properties in the PTO **912** and baggage **924** are possible to regulate the blending between focus and context. Since the PTO **912** comprises a small fraction of the total number of voxels in the scanned image, the increased time taken due to a higher sampling is minimal.

FIG. **9D** illustrates the PTO (focus) displayed as occluded and obstructed, along with contextual information, whereby everything in front of the focus **912** is dynamically cut away or removed, thus creating a virtual vision channel to visualize deep-seated PTOs. FIG. **9E** is another illustration of the focus **912** displayed as un-occluded or unhindered, along with contextual information, whereby everything in front of the focus is dynamically cut away or removed, thus creating a virtual vision channel to visualize deep-seated PTOs. Referring both to FIGS. **9D** and **9E**, a virtual cut-away surface, or view-dependent poly-plane **920** is used to clip away anything in front of the ROI **908**, as defined by the bounding box. This provides an unhindered vision channel for the focus while retaining the rest of the contextual imagery beside and behind it. Thus, as shown in FIG. **9E**, the poly-plane **920** was used to clip off a top portion of the items on top of or in front of the PTO **912**.

FIG. **9F** is an illustration wherein a user begins defining a region of interest **930** (FIGS. **9G**, **9H**) around an object **912** such that it can be isolated in the same manner as a PTO. FIG. **9G** shows the region of interest **930** being “grown” or “expanded” around the object **912** as a result of user manipulation of the display. FIG. **9H** illustrates the region of interest **930** being further defined and moved such that it can be isolated in the same manner as a PTO. Thus, the same operations performed on the PTO as described above can be performed to isolate any object of interest within a baggage or other parcel.

FIG. **10** illustrates a flow chart for an exemplary process adapted by some embodiments of the present specification to scan an object. At **1002**, a scanning system is used to transport an object under inspection through the scanning volume. In some embodiments, the scanning system is deployed for security purposes for scanning of objects. In some embodiments, the scanning system is deployed for medical purposes for scanning living beings. At **1004**, the object under inspection is irradiated with X-rays generated by a multi-focus X-ray source. The X-ray source has a plurality of X-ray source points arranged around the scanning volume. The X-ray sources are provided and arranged according to any of the known scanning methods including and not limited to CT scanners and RTT scanners. At **1006**, the irradiated X-rays transmitted through the object under inspection are detected using a detector array positioned between the X-ray source and the scanning volume, wherein the detector array has a plurality of multi-energy detector modules arranged around the scanning volume. At **1008**, reconstructed image data of the object under inspection is

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analyzed to identify a PTO within the object. In some embodiments, the process of analyzing comprises semantic classification of regions within the image. The semantic classification may identify regions with voxels including PTOs and other regions that do not include PTOs. At **1010**, the PTO is displayed in isolation along with spatial and contextual information relative to the object under inspection, in accordance with the various embodiments described in context of FIGS. **5A** to **5F**, FIG. **7**, FIGS. **8A**, **8B** and FIGS. **9A** to **9H**.

Aviation security screeners or operators are required to review and analyze scan images of baggage before clearing a piece of baggage that is to be loaded onto an airplane for a flight. The screeners are faced with the challenge of reviewing many bags within a short period of time. In a typical operation a bag that is rejected by a first level or stage screener is sent for further analysis to a second level or stage screener. The second level screener may spend more time for his/her analysis before making a final decision on the bag. If the bag is rejected by the second level screener then the bag is sent for a physical examination or hand search of the bag. Any observations or comments that can be communicated by the first level screener is valuable to the second level screener and helps him to make a decision on the bag within the limited time available.

Therefore, in accordance with some embodiments, the present specification recognizes the fact that an audible recording of the comments, where the first operator can speak his thoughts and observations in association with a scan image, are very useful. While the first operator records his observation(s) using a microphone, his hands are free without having to type in his comments using a keyboard and he can actively manipulate the scan image for quick analysis of the scan data. Likewise, the next level operator who is viewing the scan image can also manipulate the scan image while simultaneously listening to the recorded comments of the first level operator using a headphone and thus saving valuable time without having to read details or toggle between screens. The recorded audio comments associated with the scan image can also be used for audit purposes if required for TIP (Threat Image Projection) performance analysis or in case of an investigation.

In the following description, it should be noted that the system enables an operator to interactively record notes/comments or listen to notes/comments left by another operator, by clicking on it from the display or otherwise interacting with the scan to select the record functionality. In alternative methods of interaction, the user may touch any button provided on a GUI, which may be implemented on a touch screen, so as to select the appropriate action.

FIGS. **11A** through **11F** illustrate an exemplary sequence of steps that enable at least two security screeners or operators to record audio comments or observations related to a scan image of a bag, in accordance with some embodiments of the present specification. FIG. **12** is a flowchart describing the process of FIGS. **11A** through **11F**, in accordance with some embodiments of the present specification. Referring simultaneously to FIGS. **11A** through **11F** and FIG. **12**, an exemplary sequence of steps that enable at least two security screeners or operators to record audio comments or observations related to a scan image of a bag, are now described. At step **1202**, a first level screener opens a scan image of a baggage for review. The scan image is shown on a display, such as the display **300** of FIG. **3**. In some embodiments, as shown in FIG. **11A**, the display also provides a comments GUI (Graphical User Interface) window or tab **1105** that includes a text box **1107** and a record

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button or icon **1110**. In some embodiments, the first level screener may enter textual comments **1106** in the text box **1107**.

At step **1204**, and referring to FIG. **11B**, the first level screener clicks (or touch enables) the record button **1110** (FIG. **11A**) and speaks into a microphone to begin recording his audio comments and observations while simultaneously reviewing the scan image. The record button **1110** toggles into a stop button **1111** (FIG. **11B**) which when clicked (or touched) enables the screener to stop recording. In some embodiments, there is a predefined limit in terms of the amount of audio file storage available to the screener for recording. In some embodiments, the predefined audio file storage limit is provided in terms of a total time of recording. However, in alternate embodiments, the predefined audio file storage limit may be provided in terms of storage size. In some embodiments, an audio recording progress bar **1115** shows the amount of audio file storage being used by the screener during recording and the amount of audio file storage available for recording. A headset (comprising a microphone and earphones) is provided to all screeners to enable them to record their audio comments and listen to audio clips recorded by other screeners.

At step **1206**, and referring to FIG. **11C**, the first level screener finishes audio recording of his comments by clicking on the stop button **1111** (FIG. **11B**). The recorded audio information is associated with the scan image and saved as a file **1117** along with a date and time stamp as well as the first level screener's ID (Identification). The saved audio file **1117** appears as an icon that can be clicked to play the file **1117**. In some embodiments, the first level screener has an option to delete the file **1117** (by clicking a delete button **1118** associated with the file **1117**) before making a decision on the bag. However, once the first level makes a decision on the bag, the file **1117** is automatically saved and cannot be deleted by the first level screener. Note that the stop button **1111** of FIG. **11B** toggles back to the record button **1110** in FIG. **11C**.

Assuming that the first level screener makes a decision that the bag contains a PTO and refers the decision to a second level screener then, at step **1208**, the second level screener opens the scan image of the baggage for review. At step **1210**, and referring to FIG. **11D**, the second level screener examines the scan image while simultaneously clicking (or touching) on the first level operator's saved audio comments file **1117** for replay to help with analysis of the scan image and thus, help arrive at a final decision. The first level screener's textual comments **1106** are also visible to the second level screener. The second level screener can also see the ID of the previous screener (that is, the first level screener) and the time at which the audio comment was recorded. As shown in FIG. **11E**, the second level screener can choose to add textual comments **1120** to the text box **1107** so that these comments **1120** appear sequentially below the first level screener's textual comments **1106**. In FIG. **11E**, is it shown that the second level screener is also listening to the saved audio comments file **1117**.

At step **1212**, and referring to FIG. **11F**, the second level screener clicks on the record button **1110** and speaks into a microphone to begin recording audio comments and observations (to provide further details on the image observation) while continuing to review and manipulate the scan image. Once recorded, the second level screener's recorded audio information is also associated with the scan image and saved as a file **1125** along with a date and time stamp as well as

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with the second level screener's ID. The saved audio file **1125** appears as an icon next to the first level screener's audio file **1117**.

At step **1214**, the same scan image is viewed by subsequent level screeners wherein the subsequent level screeners are able to listen to each of the previous screener's audio recording with the date and time stamp and screener ID, record their audio comments and observations on the scan image and save the generated audio file in association with the scan image (along with their respective screener IDs and date and time stamps of generating the audio files).

In various embodiments, the audio comments and observations are useful for training purposes where the logic behind a screener's decision is analyzed. Thus, a scan image may be reviewed along with the associated one or more audio comments to understand the rationale behind the decisions made by corresponding one or more screeners. The audio comments can be used for performance analysis of the screeners. The audio comments can also be used for coaching new screeners on how to use or implement the audio comments.

While reviewing scan images for security purposes screeners are often presented with images wherein the threat voxels are highlighted in a predefined color or mask such as, for example, red color for PTOs and blue color for metallic objects. The voxels or areas that are to be visually highlighted are determined by a threat detection algorithm that processes the scan images. The highlighted mask or surfaces help the screener or operator to focus on the potential threat objects so that he can provide a decision on the threat quickly. The mask, however, also hides the surface and original image of the object. The present specification recognizes that it would be advantageous for the screener to be able to see beyond the mask so that the potential threat object is revealed.

Accordingly, in some embodiments, the screener is enabled to modulate the mask intensity. That is, the transparency of the mask can be increased or decreased to reveal and display the original object thereby providing a convenient and useful feature for the screener to study and examine a potential threat object. Thus the operator can see the original image of the object and also quickly see the areas or regions of the object that have triggered the threat detection algorithm as indicated by red highlighted areas.

FIGS. **13A** through **13E** illustrate an exemplary sequence of steps that enable a security screener to modulate the mask intensity or blend factor of highlighted one or more PTOs in a scan image of a bag, in accordance with some embodiments of the present specification. FIG. **14** is a flowchart describing the process of FIGS. **13A** through **13E**, in accordance with some embodiments of the present specification. Referring simultaneously to FIGS. **13A** through **13E** and FIG. **14**, an exemplary sequence of steps that enable a screener to control the intensity of mask used to highlight PTOs in a scan image of a bag, are now described. At step **1402**, and referring to FIG. **13A**, the screener opens a scan image of a baggage **1305** for review, in a display **1300**, wherein one or more PTOs **1310** are displayed with a colored mask such as, for example, red. In accordance with some embodiments, the display **1300** includes a slice view **1302** of the scanned baggage **1305**, shown on the left side of the display **1300** and a 3D view **1306** shown on the right side. The scanned image of the baggage **1305** includes, as an illustration, three PTOs **1310** highlighted within a colored bounding box **1315**. The display **1300** includes a menu **1316** with buttons or icons that may be used by the screener to interface with the scan images shown in the display **1300**. In

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accordance with some embodiments, the menu **1316** includes a mask control slider **1320** that can be manipulated (upwards or downwards) by the screener to increase or decrease the intensity or blend factor of the red colored mask of the PTOs **1310**. In FIG. **13A**, the mask control slider **1320** is at its top most position meaning thereby that the intensity of the red mask is highest and the mask layer is almost opaque.

At step **1404**, and referring to FIG. **13B**, the screener modulates the mask intensity or blend factor by manipulating the slider **1320**. As shown, the slider **1320** is moved downwards thereby decreasing the mask intensity or blend factor such that the mask color is decreased and more of the original threat objects **1310** are displayed. In other words, the PTOs **1310** are now displayed with their mask intensities or blend factors reduced.

At step **1406**, and referring to FIG. **13C**, the screener continues to modulate the mask intensity or blend factor using the slider **1320**. As shown, the slider **1320** is moved further downwards thereby decreasing the mask intensity or blend factor further such that the mask color is further reduced to reveal more of the original threat objects **1310**. The mask layer is now more transparent (compared to that in FIG. **13B**) showing the regions and/or surfaces on the PTOs **1310** that are more likely to be the threat.

At step **1408**, and referring to FIG. **13D**, the screener moves the slider **1320** all the way down to completely turn the masking off. As a result, the PTOs **1310** are displayed in their original form with no mask layer on them. Consequently, the screener can have a clear visibility of the objects **1310** in their respective original forms. Thus, in this step, the masking intensity or blend factor is completely removed to reveal just the underlying objects. At step **1410**, and referring to FIG. **13E**, the screener moves the slider **1320** all the way up to turn the masking on again and highlight the PTOs **1310** in their red mask/color. In other words, the screener re-instates the mask with the blend factor at the pre-set value so that the threat detection algorithm results are completely overlaid on the objects and the threat regions **1310** are once again displayed with full intensity.

It should be appreciated that using the mask intensity or blend factor modulation method of FIG. **14**, a screener can easily correlate the threat detected areas with the original object to help him with the analysis and arrive at a decision on the object quickly.

Embodiments of the present specification provide advances in state of the art 3D visualization thereby enabling better appreciation and instant visual evaluation of objects in a scanned image. The methods described herein enable intuitive and instant interaction for a Transport and Security Administrator (TSA) operator to analyze PTOs with surrounding contextual information thereby increasing the efficiency and reducing the analysis time for the screener operator. Embodiments of the present specification also facilitate an ergonomic way to manipulate the images when implemented on a touch screen to make the interaction more user friendly and easier to use.

The above examples are merely illustrative of the many applications of the system and method of present specification. Although only a few embodiments of the present invention have been described herein, it should be understood that the present invention might be embodied in many other specific forms without departing from the spirit or scope of the invention. Therefore, the present examples and embodiments are to be considered as illustrative and not restrictive, and the invention may be modified within the scope of the appended claims.

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We claim:

1. A method of enabling at least a first operator and a second operator to annotate a scan image of an object, the method comprising:

presenting a first graphical interface to the first operator, wherein the first graphical interface comprises a first icon;

upon receiving an indication that the first icon has been activated on the first graphical interface, initiating a first audio recording and changing the first icon into a second icon;

upon receiving an indication that the second icon has been activated, terminating the first audio recording and generating a first audio file of the first audio recording; displaying a third icon on the first graphical interface, wherein the third icon is indicative of the first audio file;

presenting a second graphical interface to the second operator, wherein the second graphical interface comprises the first icon and the third icon;

upon receiving an indication that the third icon has been activated, playing the first audio recording;

upon receiving an indication that the first icon has been activated on the second graphical interface, initiating a second audio recording and changing the first icon into the second icon;

upon receiving an indication that the second icon has been activated, terminating the second audio recording and generating a second audio file of the second audio recording; and

displaying a fourth icon on the second graphical interface, wherein the fourth icon is indicative of the second audio file.

2. The method of claim 1, wherein the first audio file is associated with an identification of the first operator.

3. The method of claim 2, wherein the second audio file is associated with an identification of the second operator.

4. The method of claim 2, wherein the second graphical interface is further configured to display the identification of the first operator.

5. The method of claim 1, wherein the first audio file is associated with a first date and a first time of the first audio recording.

6. The method of claim 5, wherein the second audio file is associated with a second date and a second time of the second audio recording.

7. The method of claim 1, further comprising permitting the first operator to visually maneuver the scan image in the first graphical interface during the first audio recording.

8. The method of claim 1, wherein the first graphical interface comprises an input configured to receive first textual comments from the first operator.

9. The method of claim 8, wherein the second graphical interface is further configured to display the first textual comments from the first operator.

10. The method of claim 9, wherein the second graphical interface comprises an input configured to receive second textual comments from the second operator.

11. The method of claim 10, wherein the second graphical interface is configured to display the second textual comments sequential to the first textual comments.

12. The method of claim 1, wherein the first graphical interface comprises a first audio progress bar configured to represent a remaining amount of storage available for the first audio recording.

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13. The method of claim 12, wherein the second graphical interface comprises a second audio progress bar configured to represent a remaining amount of storage available for the second audio recording.

14. The method of claim 1, wherein the first icon is indicative of a record button and wherein the second icon is indicative of a stop record button.

15. A method of enabling at least a first operator and a second operator to annotate a scan image of an object, the method comprising:

presenting a first graphical interface to the first operator, wherein the first graphical interface comprises a first icon;

upon receiving an indication that the first icon has been activated, initiating a first audio recording and changing the first icon into a second icon;

enabling the first operator to visually maneuver the scan image in the first graphical interface during the first audio recording;

upon receiving an indication that the second icon has been activated, terminating the first audio recording and generating a first audio file of the first audio recording; displaying a third icon on the first graphical interface, wherein the third icon is indicative of the first audio file;

permitting the first audio file to be deleted, provided the object has not been cleared and not permitting the first audio file to be deleted if the object has been cleared;

presenting a second graphical interface to the second operator, wherein the second graphical interface comprises the first icon and the third icon;

upon receiving an indication that the third icon has been activated, playing the first audio recording;

upon receiving an indication that the first icon has been activated, initiating a second audio recording and changing the first icon into the second icon;

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enabling the second operator to visually maneuver the scan image in the second graphical interface during the second audio recording;

upon receiving an indication that the second icon has been activated, terminating the second audio recording and generating a second audio file of the second audio recording; and

displaying a fourth icon on the second graphical interface, wherein the fourth icon is indicative of the second audio file.

16. The method of claim 15, wherein the first audio file is associated with an identification of the first operator and the second audio file is associated with an identification of the second operator.

17. The method of claim 15, wherein the first graphical interface comprises an input configured to receive first textual comments from the first operator and wherein the second graphical interface is further configured to display the first textual comments from the first operator.

18. The method of claim 17, wherein the second graphical interface comprises an input configured to receive second textual comments from the second operator and wherein the second graphical interface is configured to display the second textual comments sequential to the first textual comments.

19. The method of claim 15, wherein at least one of the first graphical interface and the second graphical interface comprises an audio progress bar configured to represent a remaining amount of storage available for the first audio recording or the second audio recording.

20. The method of claim 15, wherein the first icon is indicative of a record button and wherein the second icon is indicative of a stop record button.

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